Medicinal chemistry of Non-steroidal anti-inflammatory drugs

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Learning objectives

- Discuss the biosynthesis of prostaglandins
- Discuss the role of chemical mediators in inflammation
- Discuss the structure and nomenclature of prostaglandins
- Discuss the binding characteristics of COX1 and COX2
- Discuss SARs of drugs



Introduction

- Non steroidal anti-inflammatory drugs are one of the more widely used groups of therapeutic drugs

- Other than caffeine or ethyl alcohol
  - Aspirin may be the most widely used drug in the world.

- An estimated 70 to 100 million prescriptions are written annually for NSAIDs
  - with over-the-counter (OTC) use accounting for an additional use that may be up to sevenfold higher.

Introduction…

- Rheumatic diseases affect more individuals than any chronic illness.
- CDC estimates that more than 40 million Americans have some form of arthritis or chronic joint disorder.
  - Osteoarthritis is the most common form of arthritis in the United States affecting about 12% of Americans between the ages of 25 and 74.

Cont…

- Rheumatoid arthritis is thought to affect well over 2 million Americans
- Juvenile arthritis affects 71,000 children under 16 years of age 61,000 of whom are females.
- Non rheumatoid osteoporosis affects
  - 24 million females (half of all women older than 45 years and 90% of all women older than 75 years) and
  - 16 million males.

Cont…

- The diseases mentioned are considered to be host defense mechanisms.
  - ✓ Inflammation is a normal and essential response to any noxious stimulus that threatens the host and may vary from a localized response to a generalized response
- The resulting inflammation can be summarized as follows:

1. Initial injury causing release of inflammatory mediators (e.g., histamine, serotonin, leukokinins, lysosomal enzymes, lymphokinins, and prostaglandins);
2. Vasodilation;
3. Increased vascular permeability and exudation;
4. Leukocyte migration, chemotaxis, and phagocytosis;
5. Proliferation of connective tissue cells

- The most common sources of chemical mediators are:-
  neutrophils, basophils, mast cells, platelets, macrophages, and lymphocytes

- The etiology of inflammatory and arthritic diseases is unresolved until now,
  - ✓ Hindering the development of new agents that are curative in nature.
- Currently available drugs relieve the symptoms of the disease but are not curative.
- The arachidonic acid pathway leading to the production of inflammatory mediators
  - ✓ Provides several targets for intervention on inflammation.

- NSAIDs are Used as:-
  - ✓Antipyretic
  - ✓Analgesic
  - ✓Anti-inflammatory and
  - ✓Uricosuric agents

NSAIDs

- The most widely used NSAIDs in practice today are drugs like:-
  - ✓ Aspirin,
  - ✓ Naproxen,
  - ✓ Ibuprofen,
  - ✓ Diclofenac, and others that inhibit the cyclooxygenase (COX) enzymes.
- The NSAIDs generally work by the same mechanism with little exception

Aspirin,

- ✓ A salicylate, is some what different from the other agents in that it has property of irreversibly inhibiting platelet homeostasis
  - works to its advantage and broadens its use in coronary artery disease prophylaxis

- Variations on the molecular structures have provided improved side effect profiles of agents used.
  - ✓ For example, although phenacetin and acetaminophen are not anti -inflammatory agents they are included under NSAIDs
    - Illustrate the point that improvement in molecular structure from phenacetin to acetaminophen helped to reduce the hepatotoxicity and risk for drug-induced hemolytic anemia with phenacetin

- Gastrointestinal (GI) bleeding and complications being the most feared consequence of taking these drugs
  - ✓ An emphasis on minimizing these risks has been one of the motivations to develop less offensive agents.
- Gastrointestinal side effects are believed to be related to
  - ✓ Indirect toxic effects (inhibiting COX enzymes) and
  - ✓ Direct toxic effects (local irritation to the GI mucosa).
- Molecular changes have been made to the various NSAIDs to produce
  - ✓ More potent compounds,
  - ✓ To reduce the direct irritant effects, and
  - ✓ To create pro-drugs that lack the direct irritant effect altogether.

NSAIDs

- Anti -inflammatory drugs may act by interfering with any one of several mechanisms
  - ✓Immunological mechanisms such as antibody production or antigen–antibody complexation,
  - ✓Activation of complement, cellular activities such as phagocytosis,
  - ✓Interference with the formation and release of the chemical mediators of inflammation, or
  - ✓Stabilization of lysosomal membranes

Role of Chemical Mediators in Inflammation

- Before 1971
  - ✓Shen proposed that the NSAIDs exert their effects by interacting with a hypothetical anti inflammatory receptor
  - ✓In 1971, Vane reported that indomethacin, aspirin, and salicylate, in this descending order of potency,
    - Inhibit the biosynthesis of prostaglandins from arachidonic acid
      - –Mostly accepted theory

Prostaglandins,Thromboxanes,Prostacyclin and Leukotrienes

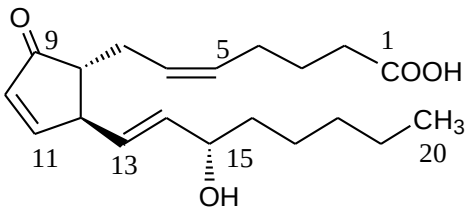
- NSAIDs are Structurally diverse agents with anti-inflammatory activity
  - ✓ Activity is attributed to their ability to inhibit *cyclooxygenase* (COX)
- Cyclooxygenase is involved in the biosynthesis of prostaglandins
- Prostaglandins are a class of *eicosanoids*
  - ✓ Eicosanoids are any product derived from arachidonic acid, a twenty carbon fatty acid
- Eicosanoids also include,
  - ✓ Thromboxanes,
  - ✓ Prostyclin, and
  - ✓ Leukotriene.

Structure of prostaglandins

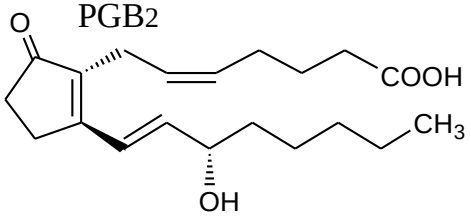
- All naturally occurring prostaglandins possess
  - ✓A 15α-hydroxy group and
  - ✓A *trans* double bond at C-13.
- Unless a double bond occurs at the C-8, C-12 positions, the two side chains
  - ✓The carboxyl-bearing chain termed the α-chain and
  - ✓The hydroxyl-bearing chain termed the β-chain are of
    - The *trans* stereochemistry depicted.

General structures of prostaglandins

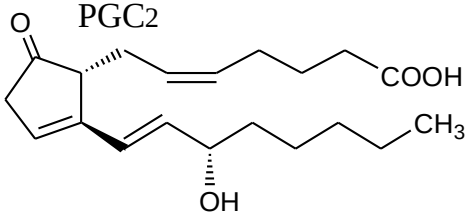
PGA2



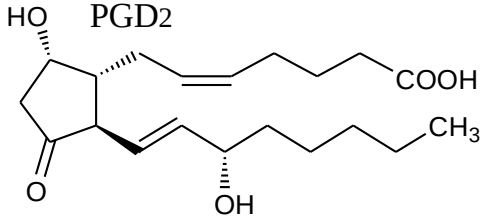
PGB2



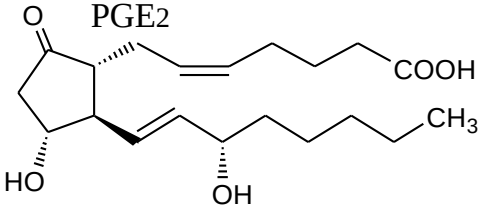
PGC2



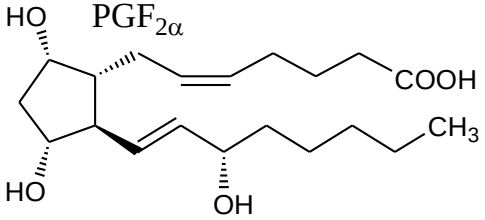
PGD2



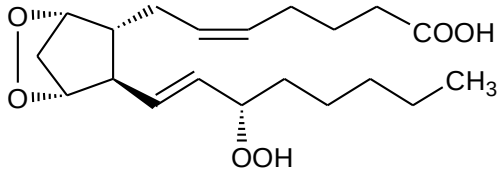
PGE2



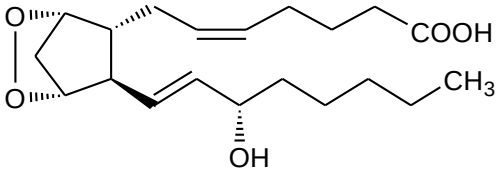
$PGF_{2\alpha}$



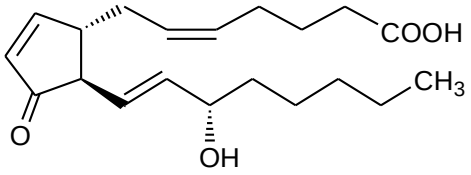
PGG2



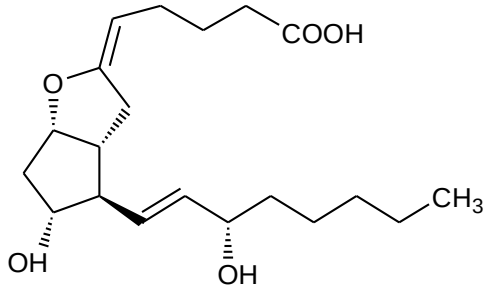
PGH2



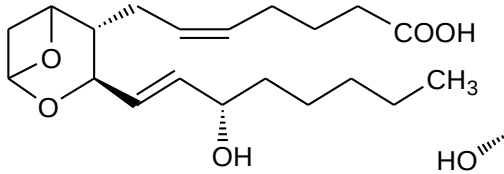
PGJ2



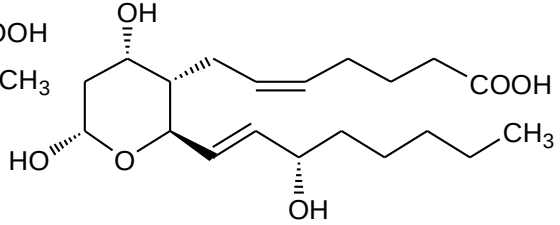
PGI2 Prostacycline



TXA2 Thromboxane A2



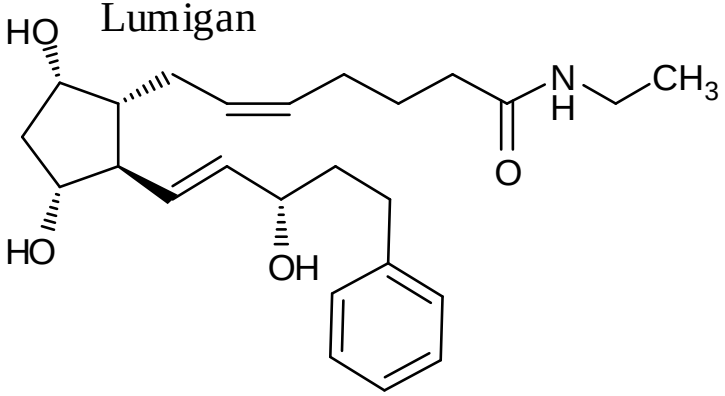
TXB2



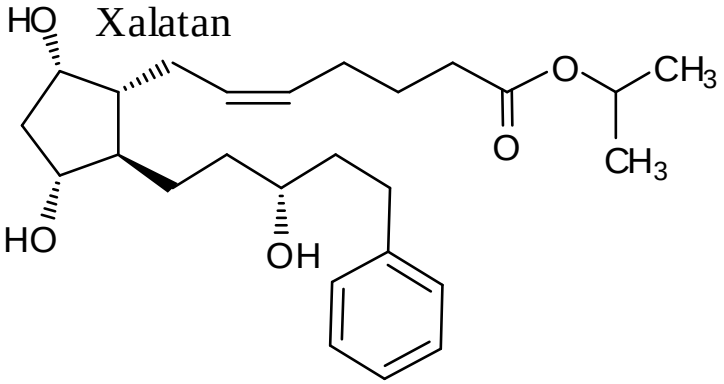
Commercial Prostanoids

A. Ocular Hypertension & Glaucoma

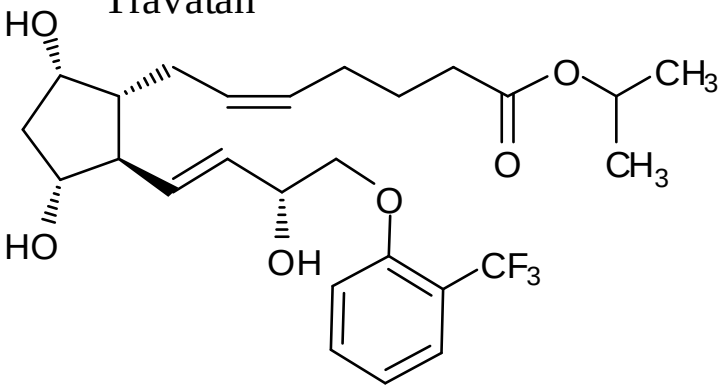
BIMATOPROST
Lumigan



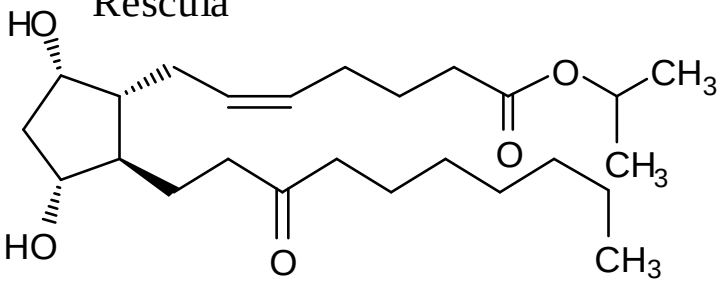
LATANOPROST
Xalatan



TRAVOPOST
Travatan



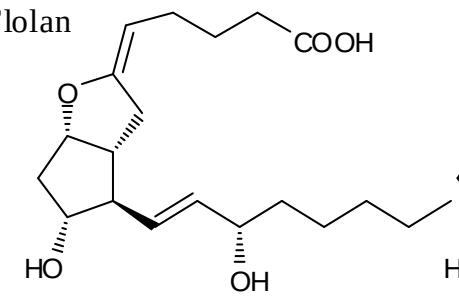
UNOPROSTONE isopropyl
Rescula



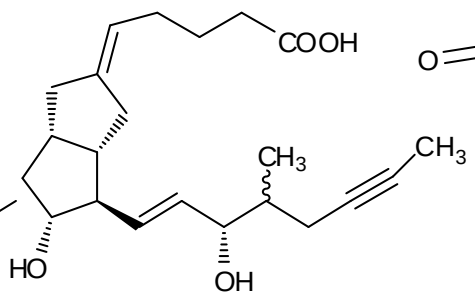
Commercial Prostanoids -II

B. Pulonary Hypertension

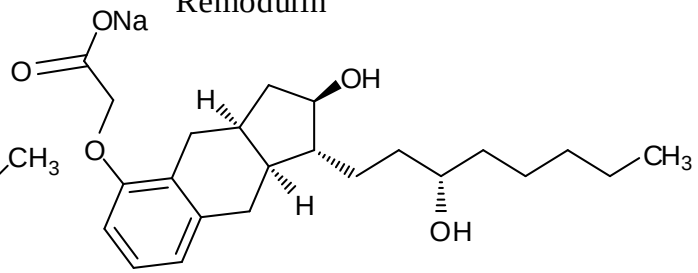
EPOPROSTENOL
PGI
Flolan



ILOPROST
Ventavis

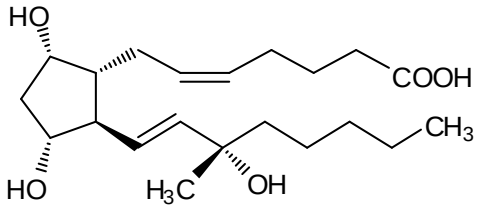


TREPROSTINIL sodium
Remodulin

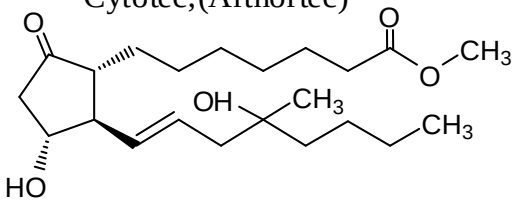


C. Other Uses

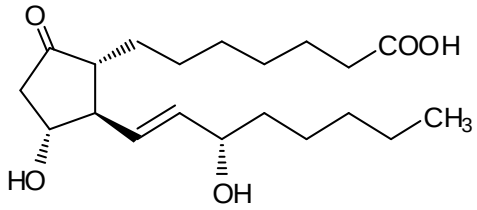
CARBOPROST
Hemabate



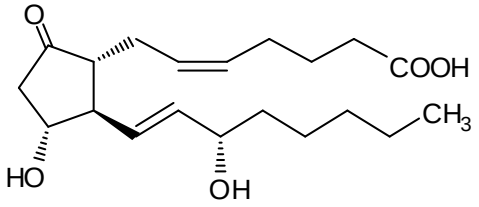
MISOPROSTOL
Cytotec,(Arthortec)



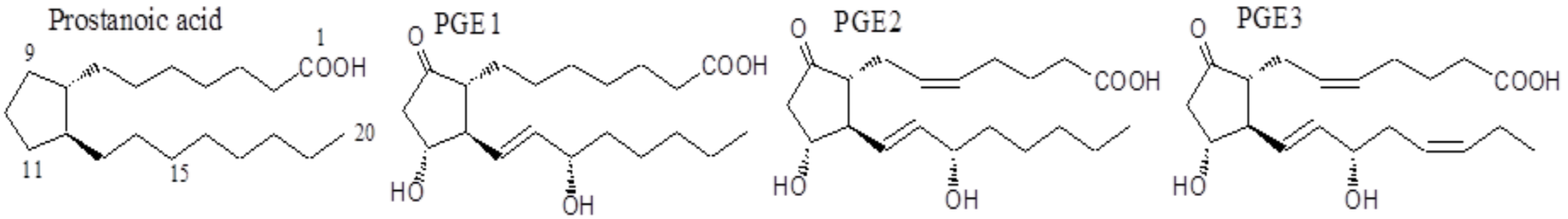
ALPROSTADIL
Prostglandin E1
Prostin VR, Caverject, Edex, Muse



DINOPROSTONE
Prostglandin E2
Prostin E2, Prepidil, Cervidil



Prostaglandins - Nomenclature



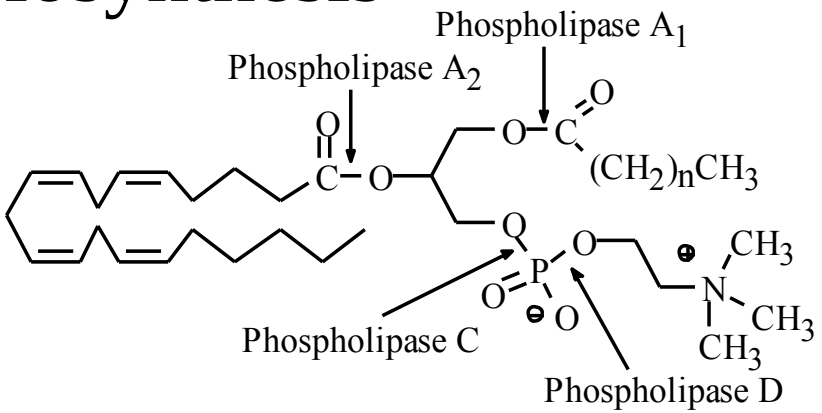
- The nomenclature is derived from the hypothetical compound prostanoic acid
- The prostaglandins are classified by the capital letters A, B, C. D, E, F, G, H, and I (e.g., PGA, PGB, and so on) depending on:-
  - ✓ The nature and stereochemistry of oxygen substituents at the 9- and 11-positions.
    - For example, members of the PGE series possess a keto function at C-9 and an α-hydroxyl group at C-11, whereas members of the PGF series possess α-hydroxyl groups at both of these positions.

Cont…

- The subscripts number 1, 2, or 3 indicates:-
  - ✓ The number of double bonds in the side chains connected to the cyclopentane ring and
  - ✓ The nature of the fatty acid precursor.
- For example:-
  - ✓ The subscript 2 indicates an additional *cis* double bond at the C-5, C-6 positions, and derived from arachidonic acid (5,8,11,14-eicosatetraenoic acid)
  - ✓ The subscript 3 indicates a third double bond of *cis* stereo-chemistry at the C-17, C-18 positions and derived from 5,8,11,14,17-eicosapentenoic acid.

Prostaglandins - Biosynthesis

- Prostaglandins are biosynthesized from Arachidonic acid



- Arachidonic acid is found esterified to a cell membrane phospholipid

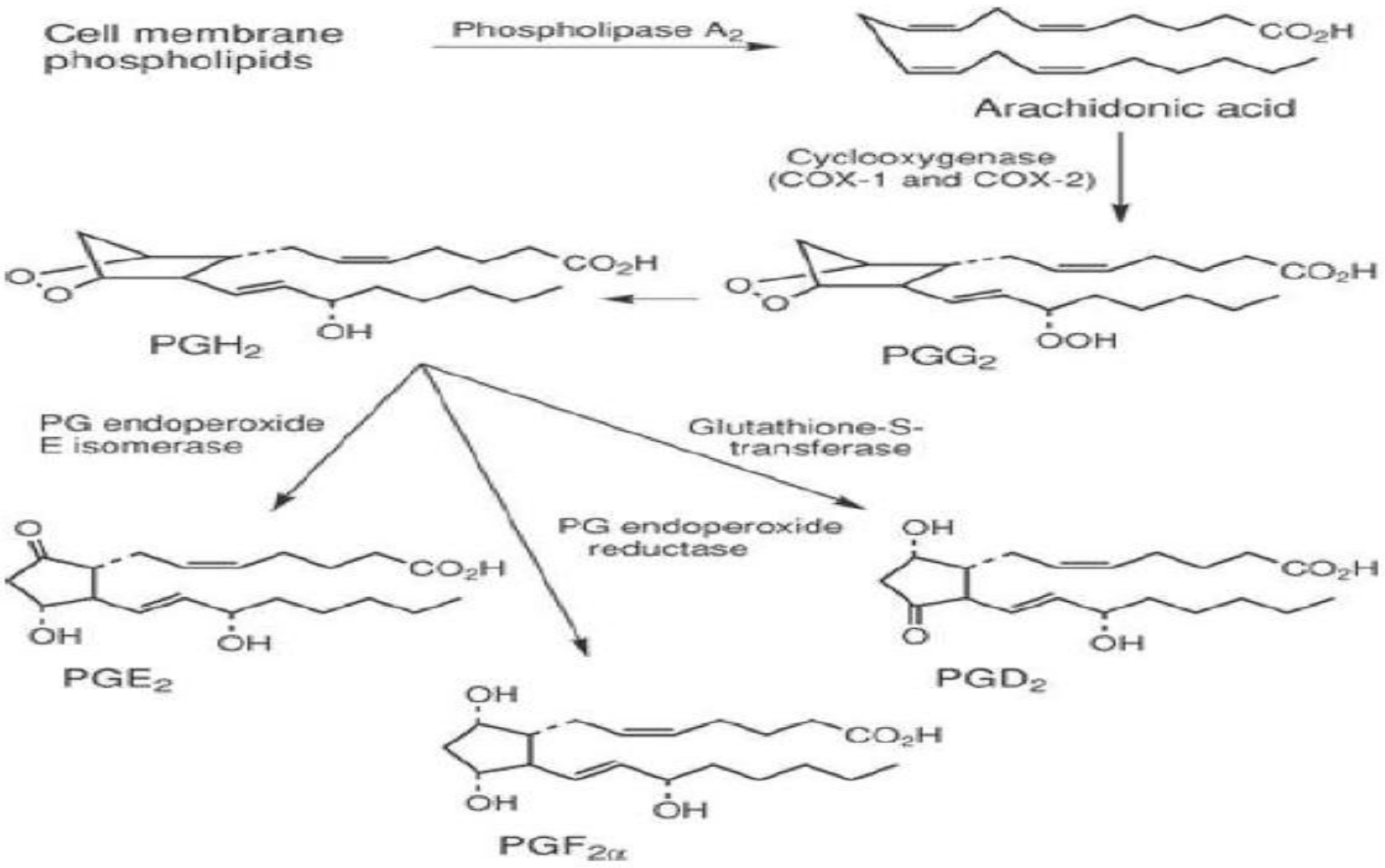
  - ✓ The concentration of free arachidonic acid is low

- The biosynthesis of prostaglandin depends primarily on its release from cellular stores by phospolipases

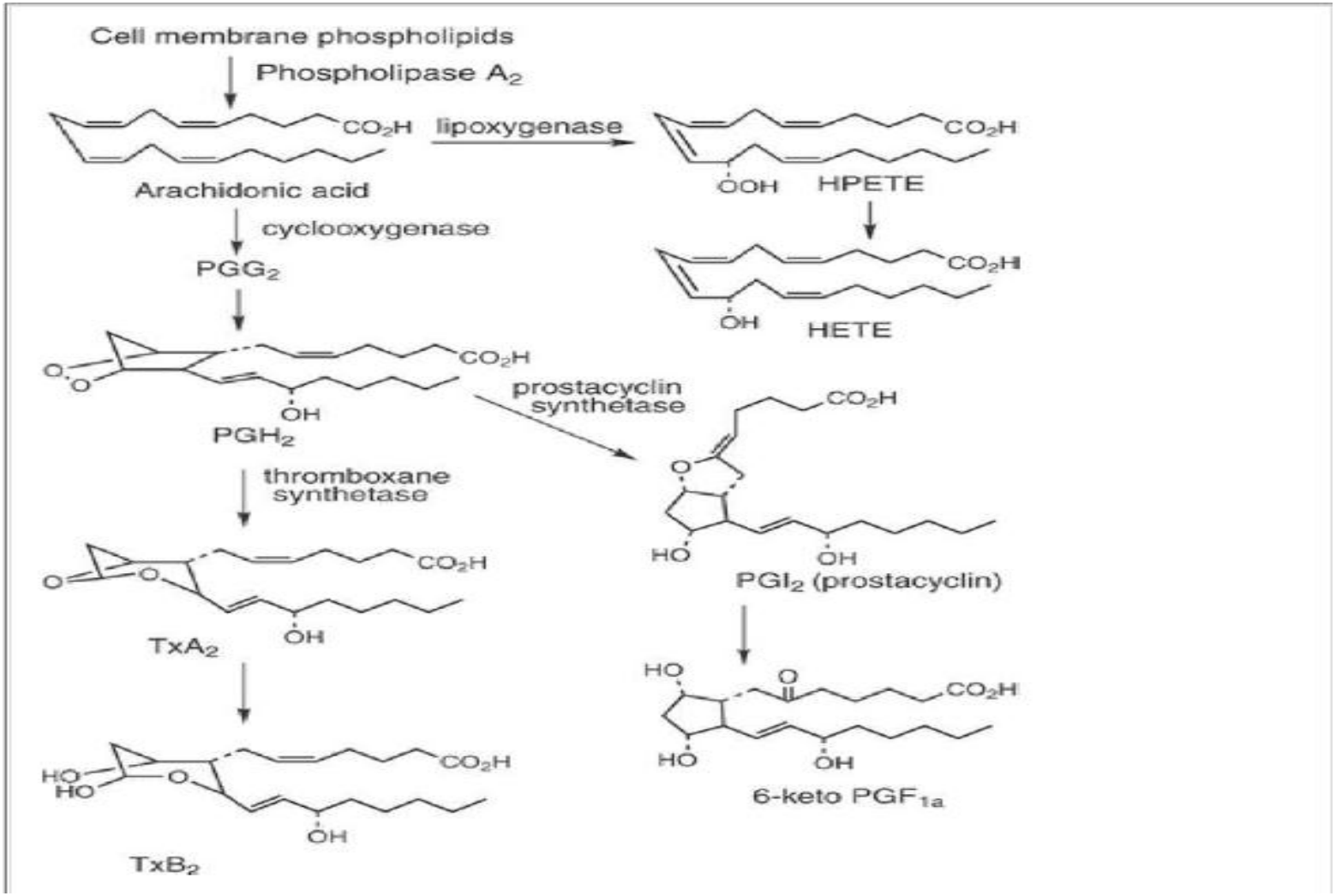
Biosynthesis cont'd…

- Membrane bound Phospholipase A2 is involved in the release of arachidonic acid from cell membrane
- Action of cyclooxygenase on arachidonic acid results in oxygenated products containing ring structures:
  - ✓ Prostaglandins,
  - ✓ Thromboxanes, and
  - ✓ Prostacyclin
- Action of various lipoxygenases result in the hydroxylated products: HPETEs, HETEs, lipoxins and leukotrienes

Biosynthesis of prostaglandins from arachidonic acid



Biosynthesis of thromboxanes, prostacyclin and leukotrienes



Protaglandin Endoperoxide Synthase

- The biosynthetic reactions are catalyzed by an enzyme complex Prostaglandin endoperoxide synthase which is located in the endoplasmic reticulum
  - ✓ It is a bifunctional enzyme with two catalytic sites adjacent but spatially distinct

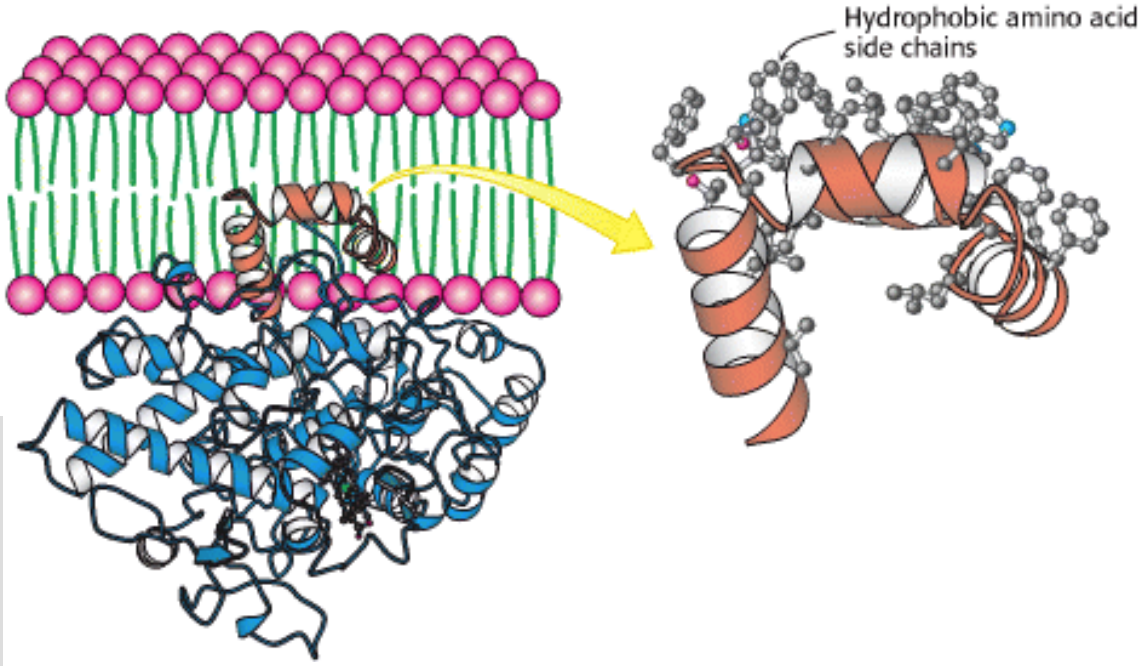
Cont…

- On one side, it has the **cyclooxygenase** active site and
- On the opposite side, it has an entirely separate **peroxidase** site
  - ✓ Activate the heme groups that participate in the cyclooxygenase reaction
- First COX oxidizes and cyclizes arachidonic acid, forming PGG2, which has an endoperoxide as well as an exoperoxide
- PGG2 then diffuse to the peroxidase catalytic site where the exoperoxide at C15 is reduced to PGH2
  - ✓ PGG2 and PGH2 are chemically unstable (t1/2 of 5 min) and are converted enzymatically by enzymes called synthetases into the other prostaglandins

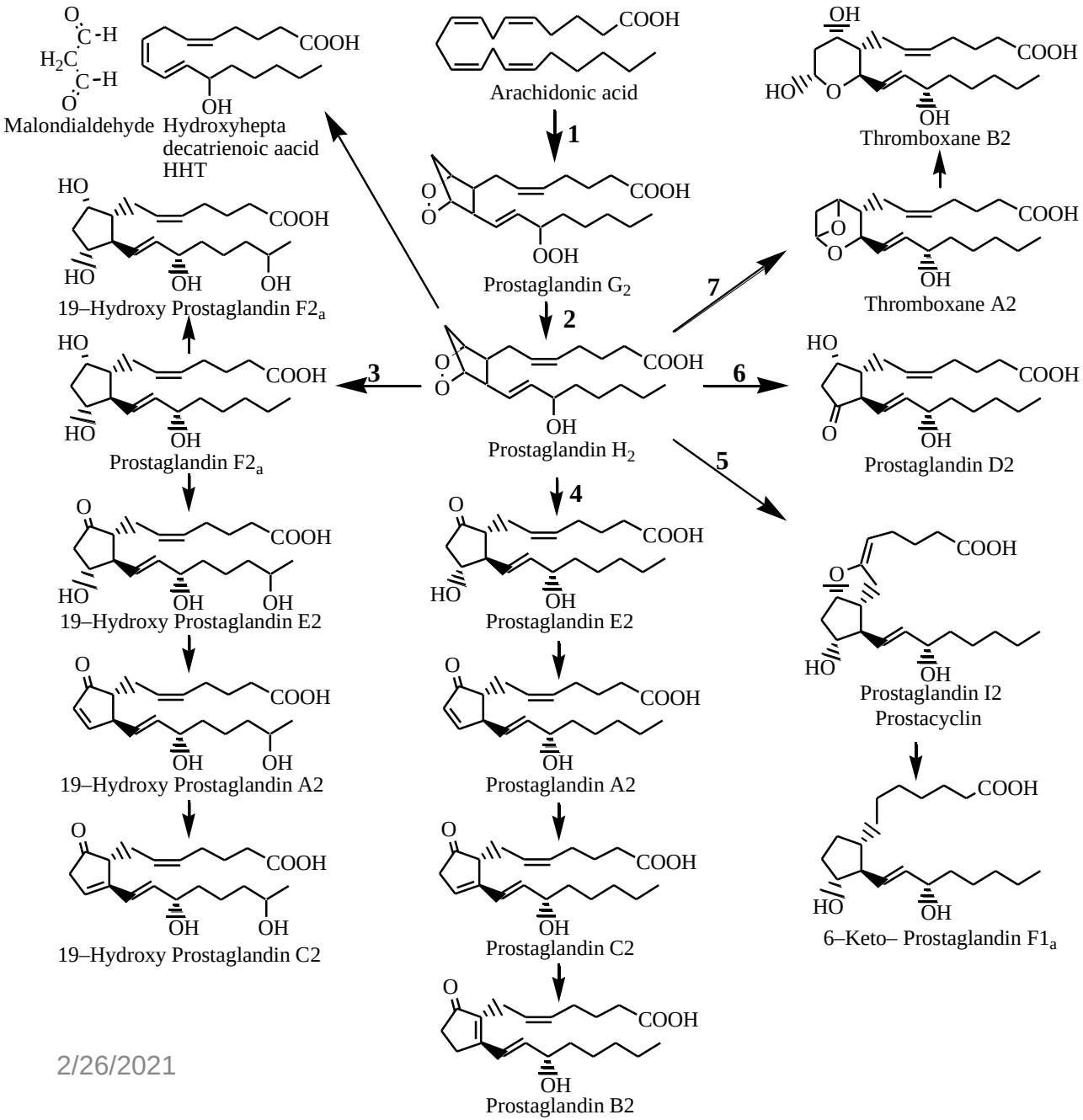
3D Structure of the Enzyme Complex

The enzyme complex is a dimer of identical subunits, so altogether, there are two cyclooxygenase active sites and two peroxidase active sites in close proximity

Each subunit has a small carbon-rich knob that anchor the complex to the membrane of the endoplasmic reticulum, shown in light blue at the bottom of the picture



The cyclooxygenase active site is buried deep within the protein, and is reachable by a tunnel that opens out in the middle of the knob. This acts like a funnel, guiding arachidonic acid out of the membrane and into the enzyme for processing



PGH2 is also converted into two unstable yet highly active thromboxanes (so named because they were first isolated from thrombocytes)

1. Cyclooxygenase
2. Peroxidase

Prostaglandin endoperoxide synthase [EC 1.14.99.1]

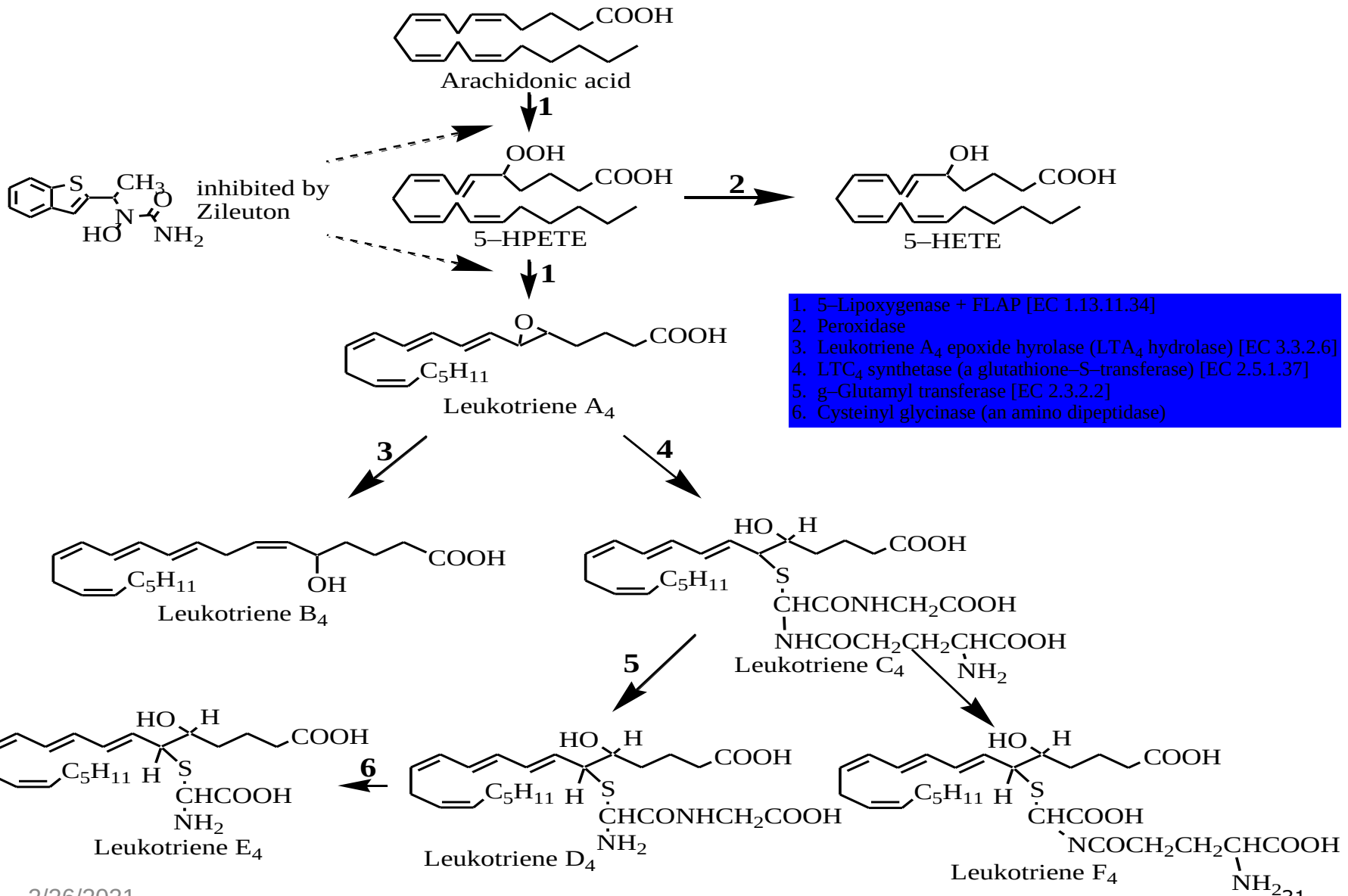
Prostaglandin endoperoxide synthase (cylcooxyenase domain)

Prostaglandin endoperoxide synthase (hydroperoxidase domain)

3. Prostaglandin F_2a synthase [EC 1.1.1.188]
4. Prostaglandin E_2 synthase [EC 5.3.99.3]
5. Prostaglandin I_2 synthase [EC 5.3.99.4]
6. Prostaglandin D_2 synthase [EC 5.3.99.2]
7. Thromboxane A_2 synthase [EC 5.3.99.5]

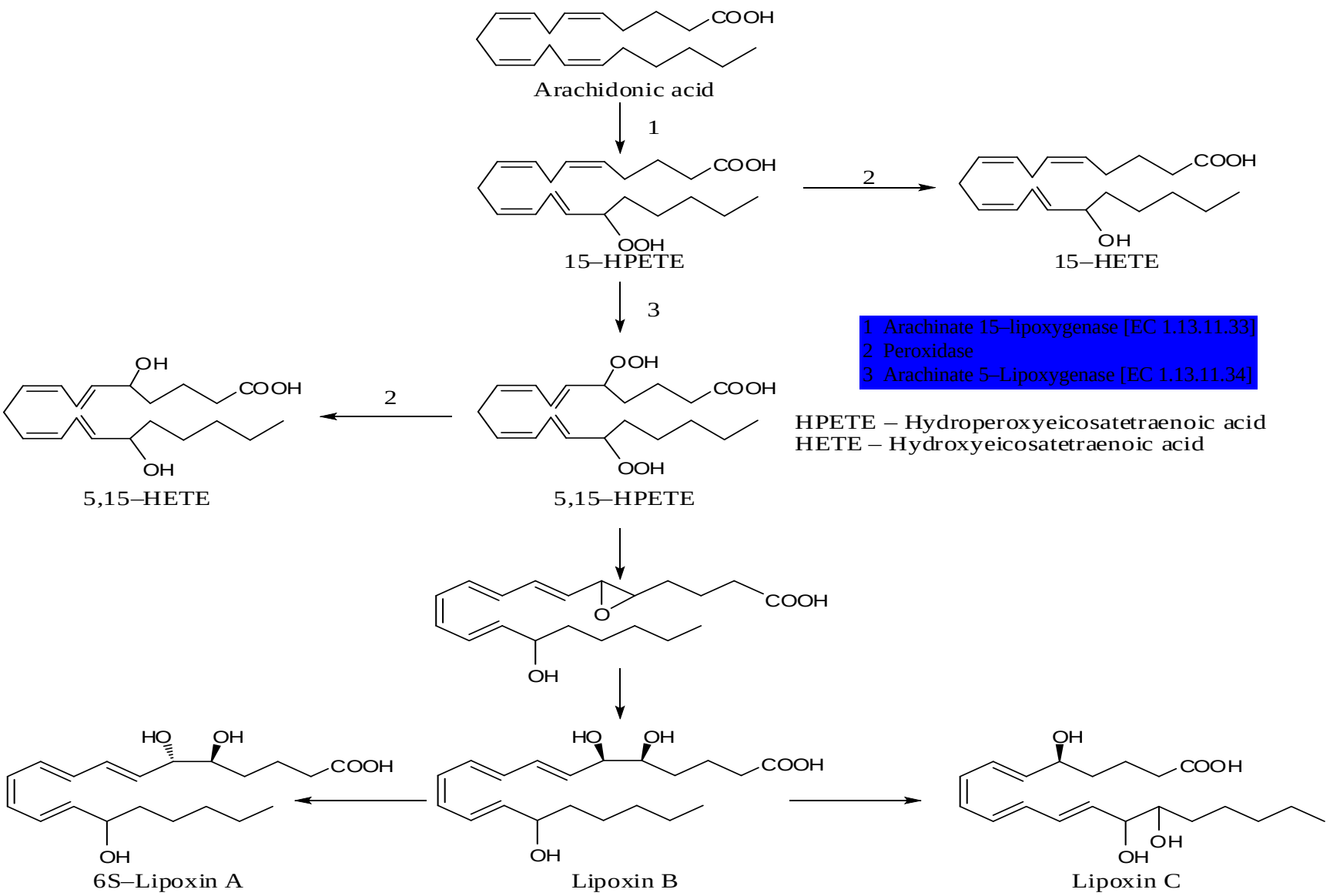
Products of 5–Lipoxygenases

Leukotriene Biosynthesis



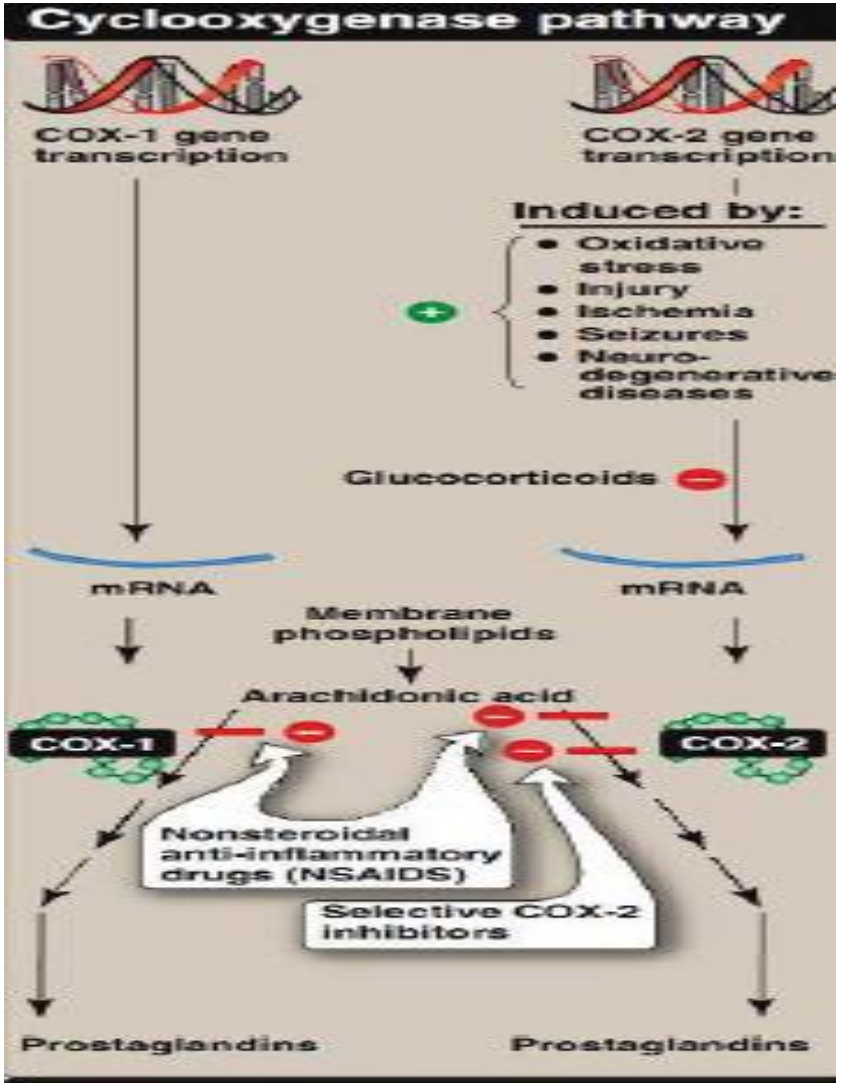
Products of 15–Lipooxygenases

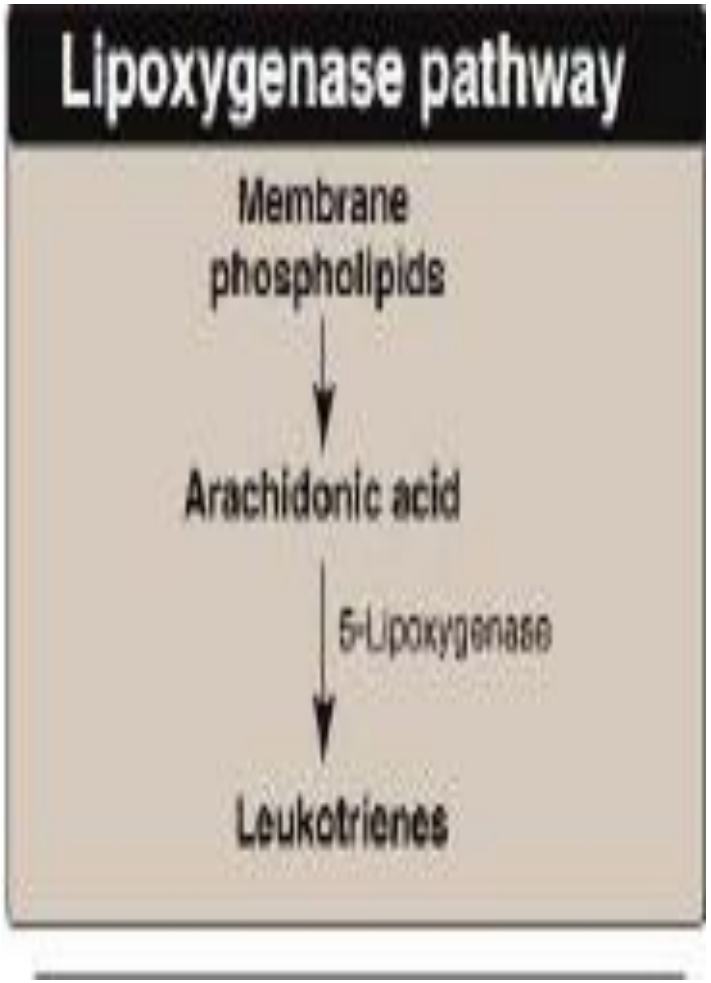
Lipoxin Biosynthesis



1 Arachinate 15–lipoxygenase [EC 1.13.11.33]
2 Peroxidase
3 Arachinate 5–Lipoxygenase [EC 1.13.11.34]

HPETE – Hydroperoxyeicosatetraenoic acid
HETE – Hydroxyeicosatetraenoic acid


Cyclooxygenase pathway
COX-1 gene transcription
COX-2 gene transcription
Induced by:
Oxidative stress
Injury
Ischemia
Seizures
Neuro-degenerative diseases
Glucocorticoids
mRNA
mRNA
Membrane phospholipids
Arachidonic acid
COX-1
COX-2
Nonsteroidal anti-inflammatory drugs (NSAIDS)
Selective COX-2 inhibitors
Prostaglandins
Prostaglandins


Lipoxygenase pathway
Membrane phospholipids
Arachidonic acid
5-Lipoxygenase
Leukotrienes

Prostaglandins - Function

- Depending on the individual Prostaglandin and the tissue they act on prostaglandins have varied actions
  - ✓ They are implicated in the inflammatory response
  - ✓ Sensitizing pain receptors to the action of other mediators
  - ✓ Play critical roles in tissue homeostasis and function
  - ✓ They have a cytoprotective role in the kidney and gastric mucosa.

Pharmacological Properties of Prostaglandins, Thromboxane, and Prostacyclin

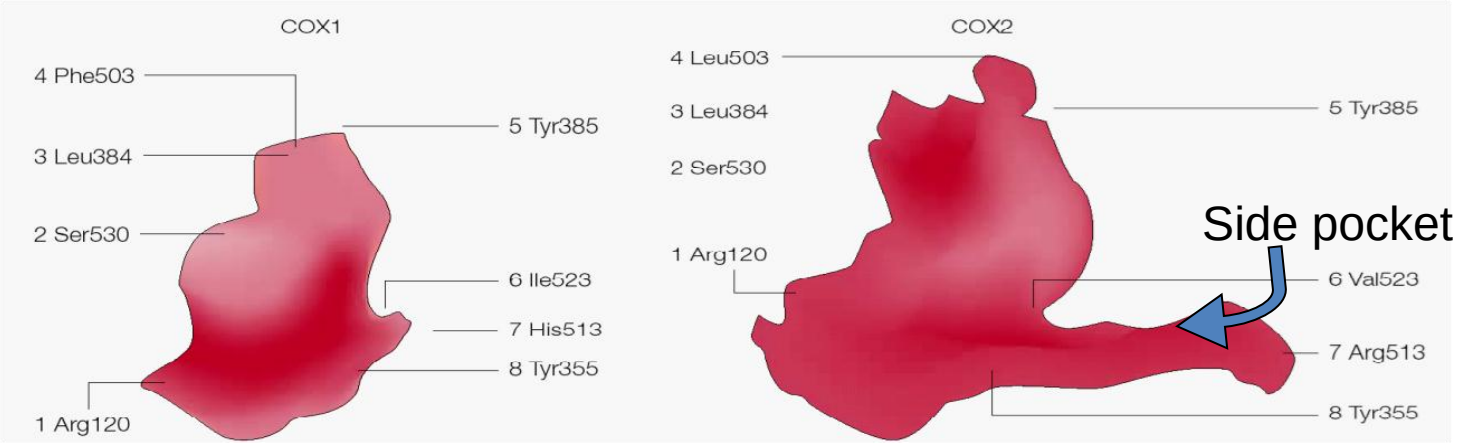
	PGE_2	$PGF_{2\alpha}$	PGI_2	TxA_2
Uterus	Oxytocic	Oxytocic		
Bronchi	dilation	constriction		
Platelets			Inhibits aggregation	Aggregation
Blood vessels	Dilation	Constriction	Dilation	Constriction

COX-1 & COX-2

- Overall structure, length and catalytic activity of both are similar
- COX-1 expression is regulated by hormonal signals involved in maintaining physiologic homeostasis
- COX-1 is expressed in all tissues
  - ✓ Importantly, COX-1 constitutively expressed in the stomach, where it is involved in mucosal defense and repair
- COX-2 expression and activity is largely responsive to adverse stimuli, such as inflammation and physiologic imbalances

COX-1 & COX-2

- COX-2 has a binding site for steroids whereas COX-1 does not
- Control of COX-2 transcription and translation is thought to be the:-
  - ✓ Primary mechanism by which steroids such as hydrocortisone and dexamethasone modulate this enzyme.
- COX-2 is constitutively expressed notably in the brain and kidney



- **In COX1** residues Arg120 &Tyr355 stabilize the anionic group present in most NSAIDs.
  - NSAID aromatic rings are accommodated in the hydrophobic channel.
  - Ser530 is the residue acetylated by aspirin.
  - The bulky Ile523
    - block inhibitor access
- **In COX2** residues Arg120, Tyr355 & Ser530 are present.
  - Val523 which allows opening of a side pocket.
    - ✓ Accommodates the sulfonamide or isoster of COX2 inhibitors.
  - They are stabilized by hydrogen bonding with Arg513.
  - Selective inhibitors of COX-2 do not bind to Arg-120

Cont…

- COX–1 provides a cytoprotective role in the stomach
  - ✓ It helps maintain the integrity of the mucosal epithelium and inhibition leads to gastric damage, hemorrhage, and ulceration

- COX–2 produces prostaglandins involved in inflammation, fever and pain

- COX–2 inhibition produces therapeutic effects and COX–1 inhibition produces unwanted side effects.
  - ✓ Unfortunately, most NSAIDS are more effective at inhibiting COX–1 than COX– 2

Classes of COX Inhibitors

- The COX inhibitors can be grouped into **four** classes based on their mechanism of action.
  1. **Irreversible inhibitors:-** Aspirin is the only known member of this group
  2. **Reversible competitive inhibitors of both COX:-** freely reversible
  3. **Slow time dependent inhibition of both COX**:-
     - ✓ They bind and induce a conformational change in the enzyme
       - It can take several seconds to minutes to reach equilibrium between the reversible and pseudo irreversible complex.
       - Bind very tightly and dissociated very slowly.
     - In vivo both mechanism 2 and 3 are essentially the same.
  4. **Selective reversible competitive inhibitors COX–2:-**these agents induce a slow conformational change in COX-2 but not in COX-1.

Cont…

- With the exception of Aspirin, all the NSAIDS are reversible competitive inhibitors

- Research suggests a role for PG in CNS transmission
  - ✓ Selective COX–2 inhibitors may modulate CNS function.
    - This is relevant for those COX–2 inhibitors that lack an acidic group and thus can easily pass the BBB

NSAIDs & COX

- The "classical" nonselective NSAIDs bind to both COX-1 and COX-2,
  - ✓ By interacting with the hydrophobic channel of the COX isoenzymes
- Aspirin, unlike other NSAIDs, irreversibly acetylates a serine residue in both COX-1 and COX-2 preventing arachidonic acid from reaching the catalytic site
  - ✓ Other nonselective NSAIDs compete directly with arachidonic acid

Cont…

- **Coxibs**, the COX-2-selective inhibitors, preferentially bind to COX-2 and inhibit it.
  - ✓ They are selective agents
- Preferential inhibition of COX-2 is thought to be due to:-
  - ✓ Additional space in the COX-2 hydrophobic channel and
  - ✓ The presence of a side pocket in the channel.
    - This side pocket can discriminate the coxibs from nonselective agents based on the different overall structures of these agents, in particular, by the presence in coxibs of specific side chains
- NSAIDs do not affect the peroxidase site

The COX Binding Site

- A cationic center and two hydrophobic areas
  - ✓The cationic site is attributed to a guanidinium group on Arginine
  - ✓The first hydrophobic area is located adjacent to the cationic center
  - ✓The second region lies under and out of the plane with the first hydrophobic area and is commonly referred to as a trough

Binding site…

- Some agents can bind only the cationic center and the first hydrophobic area

- The only way to bind both hydrophobic regions simultaneously is if
  - ✓The drug contains two aromatic ring systems that are perpendicular and not coplanar

Cont…

- Binding to the trough can enhance potency.
  - ✓ If the ring cannot fit into the trough then it bangs into the walls of the enzyme, sterically inhibiting binding
- If the two rings are separated by one or more sigma bonds
  - ✓ The two rings may assume a large number of possible conformations
    - ➢ Only few of them compliment the receptor.
      - ❖ Making rigid molecule with correct conformation gives potent drugs

SAR Summary for COX Inhibitors

1. Molecule must have an ionizable acid group and an aromatic ring system
2. A second non coplanar aromatic ring increases potency by increasing bonding interactions
3. Limiting the number of possible conformers increase potency
4. A two atom separation between the anionic charge and the aromatic ring is the optimal

Cont…

5. Increasing the distance to 3 or 4 carbons generally decreases potency

6.Introduction of a methyl at the first carbon increases potency and introduces a chiral center

- The S–isomers are the more potent isomers

8. Increasing the size of the alkyl decreases potency but incorporation of the alkyl into a heterocycle retains activity

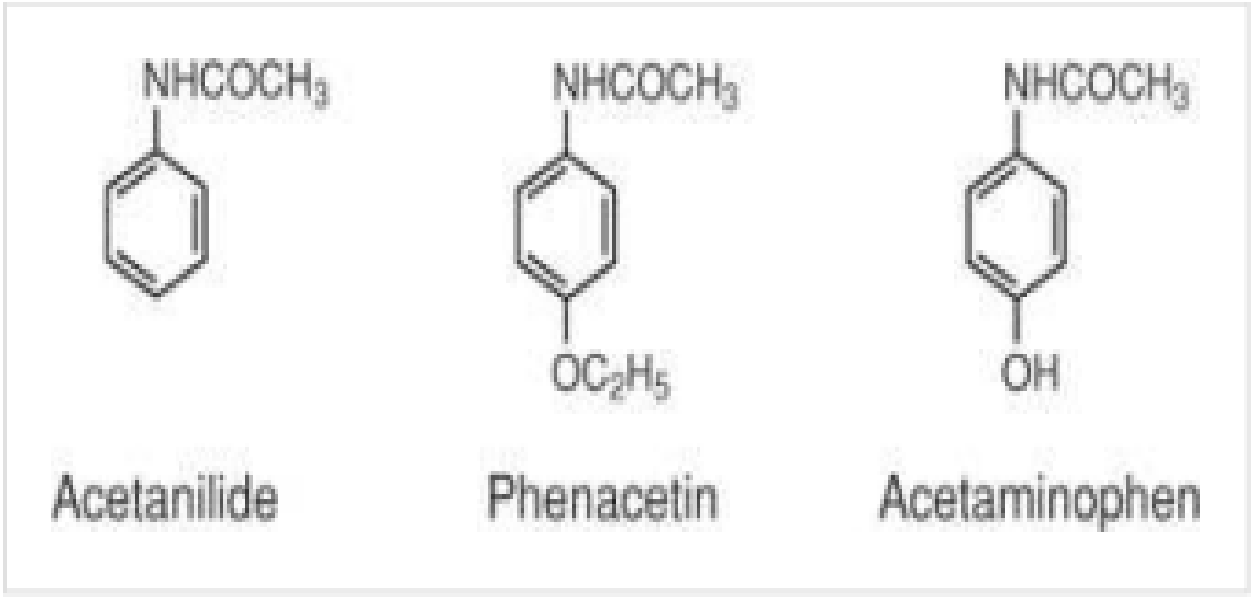
Therapeutic classification

1. Antipyretic Analgesic
2. Anti-inflammatory drugs

Antipyretic Analgesic

P-aminophenols

- Drugs included in this class possess analgesic and antipyretic actions but lack anti-inflammatory effects.



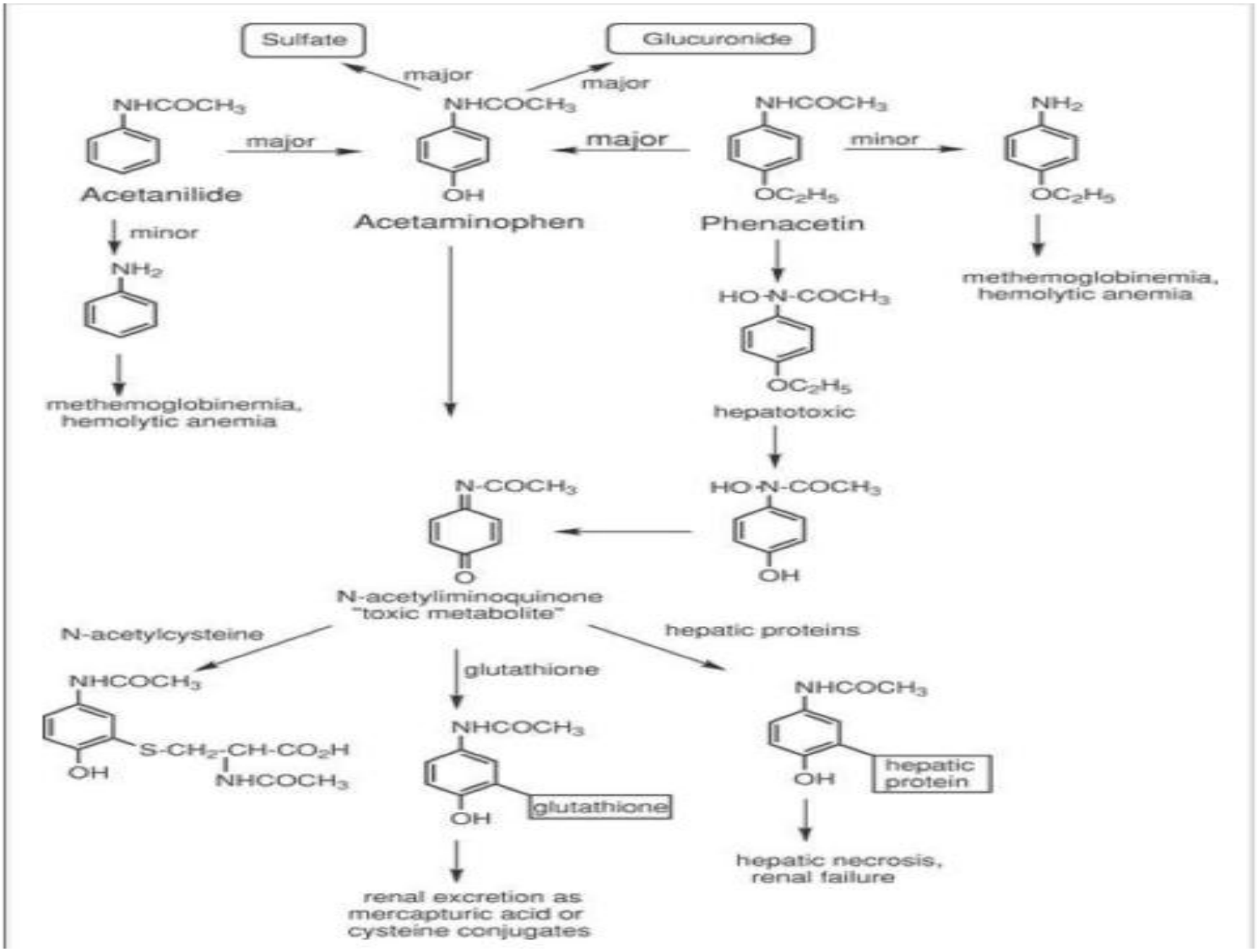
Cont…

- The first drug Acetanilide is out of market due to toxicity (both blood and liver disorders)
- Phenacetin (1887) was used for decades
  - ✓ In the 1970s it was implicated in cases of liver and nephrotoxicity and was removed from the market
- Acetaminophen is also a very old drug
  - ✓ A metabolite of both phenacetin and acetanilide,
  - ✓ Is a ‘safe drug’, producing much better tolerance and a lower incidence of gastric bleeding compared to many of the other NSAIDs
    - ✓ Because of Its apparently different mechanism of action

SAR

- Although p-aminophenol itself is too toxic for therapeutic purposes.
  - ✓ Etherification of the phenolic function with methyl or propyl groups produces:-
    - Derivatives with greater side effects than with ethyl groups
  - ✓ Substituents on the nitrogen atom that reduce basicity reduce activity
    - Unless that substituent is metabolically labile (e.g., acetyl).
  - ✓ Amides derived from aromatic acids (e.g., N-phenylbenzamide) are less active or inactive.

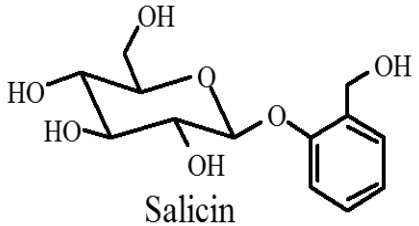
Metabolism and Toxicity



Anti-inflammatory drugs

Salicylates

- Salicylic acid was first obtained in 1838 from salicin
  - ✓ A glycoside present in most willow and poplar bark.
- In 1860, Kolbe synthesized salicylic acid from sodium phenoxide and carbon dioxide
  - ✓ Derivatives of salicylic acid began to receive medical attention shortly
  - ✓ A method that inexpensively produced large quantities.



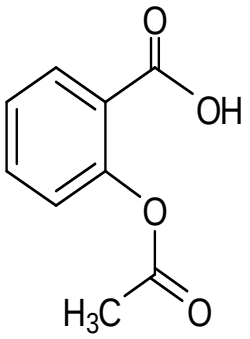
Salicin

Salicylates

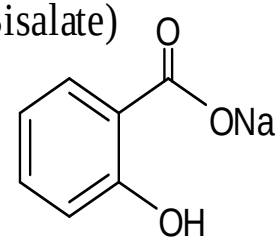
- They act as:-
  - ✓Antipyretic,
  - ✓Analgesic, and
  - ✓Anti-inflammatory properties
  - ✓Uricosoric agent
  - ✓Inhibition of platelet aggregation
- Aspirin is the only NSAID that covalently modifies cyclooxygenase by acetylating Ser -530 of COX-1 and Ser- 516 of COX- 2.

Structures of marketed derivatives of salicylic acid

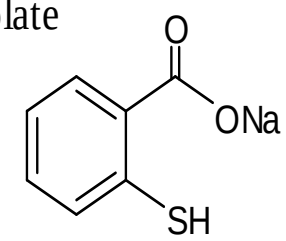
ASPIRIN



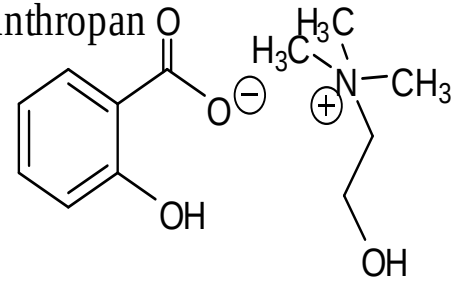
SODIUM SALICYLATE
(Bisalate)



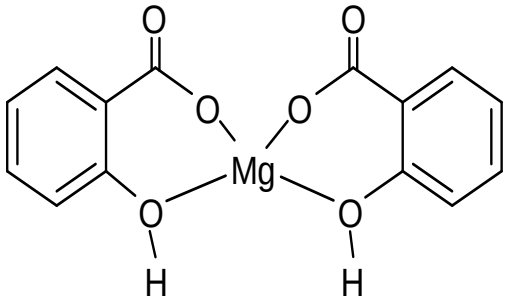
SODIUM THIOSALICYLATE
Rexolate



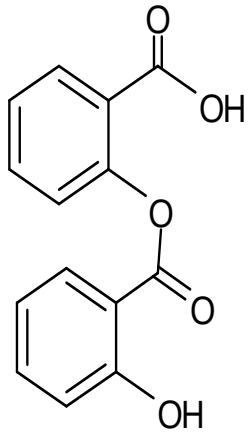
CHOLINE SALICYLATE
Anthropan



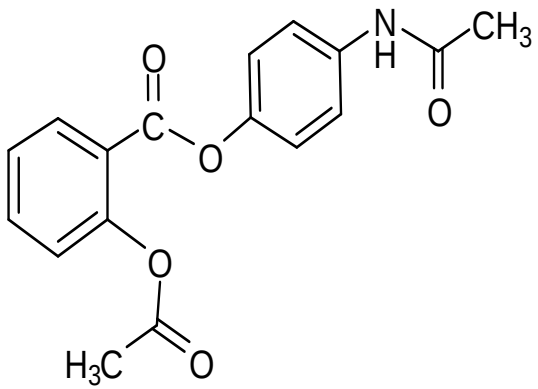
MAGNESIUM SALICYLATE
Magan, Mobidin, (Trisalate)



SALSALATE
Salflex, Disalcid



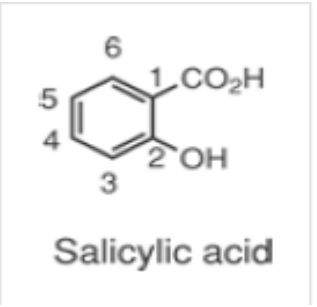
Benorylate



Mechanism of action

- A number of possible mechanisms of action have been proposed for salicylates over the years
  - ✓ Inhibition of the biosynthesis of histamine,
  - ✓ Antagonism of the actions of various kinins,
  - ✓ Inhibition of mucopolysaccharide biosynthesis,
  - ✓ Inhibition of lysosomal enzyme release, and inhibition of leukocyte accumulation
- The most widely accepted mechanism of action currently is:-
  - ✓ The ability of these drugs to inhibit the biosynthesis of prostaglandins at the cyclooxygenase stage

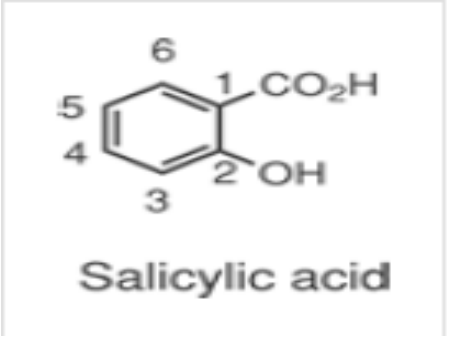
Salicylate SARs



Salicylic acid

- The active moiety appears to be salicylate anion
- The side effects of aspirin, particularly the GI effects, appear to be associated with the carboxylic acid function.
  - ✓Reducing the acidity of this group (e.g. converting to an amide, salicylamide)
    - maintains the analgesic actions of salicylic acid derivatives but eliminates the anti-inflammatory properties.

Cont…

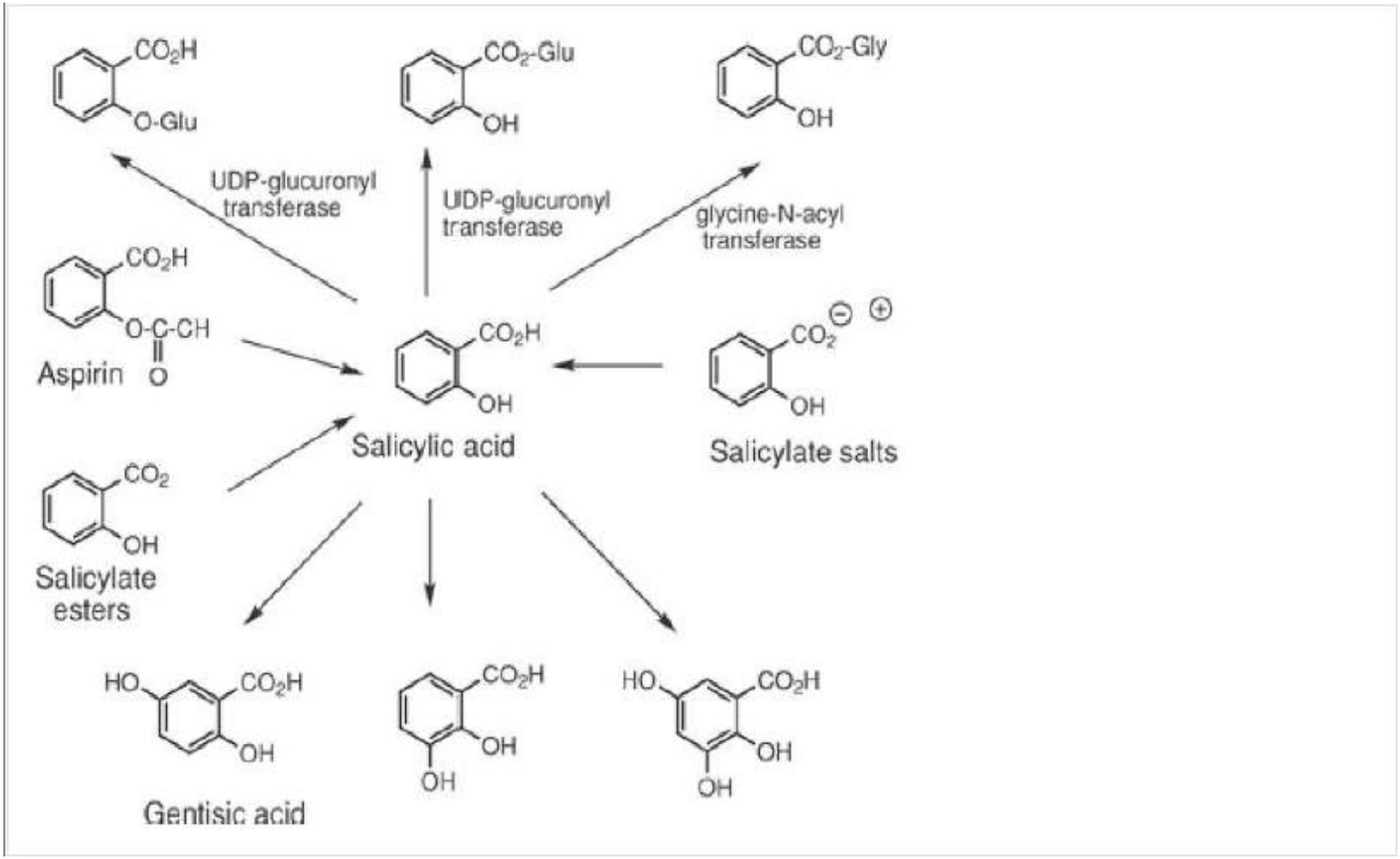


- The carboxylic group is necessary for activity and the hydroxyl group must be ortho to it.
  - ✓Placing the phenolic hydroxyl group meta or para to the carboxyl group abolishes activity.
- Substitution of halogen atoms on the aromatic ring enhances potency and toxicity.
- Substitution of aromatic rings at the 5-position of salicylic acid increases anti-inflammatory activity (e.g., diflunisal).

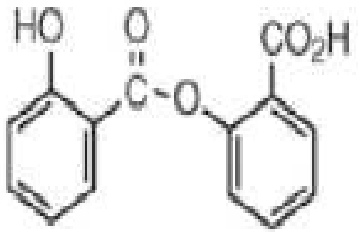
Metabolism

- The initial route of metabolism of ester and salts of salicylic acid is their conversion to salicylic acid,
  - ✓ which may be excreted in the urine as the free acid (10%) or
  - ✓ Undergo conjugation with either glycine, to produce the major metabolite salicyluric acid (75%), or
  - ✓ with Glucuronic acid, to form the glucuronide ether and ester (15%).
- In addition, small amounts of metabolites resulting from microsomal aromatic hydroxylation are found.
  - ✓ The major hydroxylation metabolite, gentisic acid, was once thought to be responsible for the anti-inflammatory actions of the salicylates,
    - But its presence in trace quantities would rule out a major role for gentisic acid,

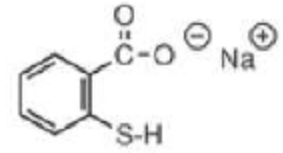
Metabolism of salicylic acid derivatives. Glu, glucuronide conjugate; Gly, glycine conjugate.



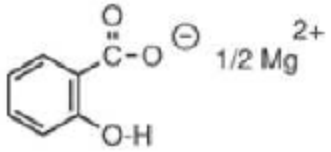
Salsalate



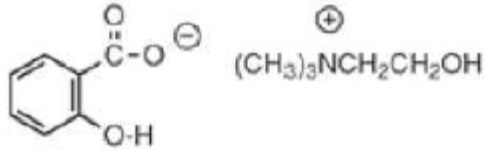
- It is prodrug esters.
- It is only half as potent as an analgesic/antipyretic as Aspirin but produces less GI irritation.
- It is a dimer of salicylic acid
- Insoluble in gastric pH but soluble in the small intestines
  - Partially hydrolyzed to two molecules of salicylic acid and absorbed.
  - Causing less gastric problems.
- It is useful in hypersensitivity to Aspirin.
  - Because produces Salicylic acid



Sodium thiosalicylate



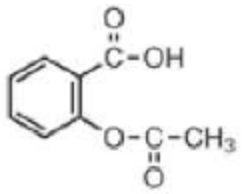
Magnesium salicylate



Choline salicylate

- **Sodium thiosalicylate** is used in rheumatic fever and acute gout and muscular pain
- **Magnesium salicylate** form stable aqueous solution and has a low incidence of GI side effects
- **Choline salicylate** is absorbed faster than Aspirin producing higher salicylate blood levels
  - Has a lower incidence of GI side effects compared with aspirin
  - Useful in treating juvenile rheumatoid arthritis

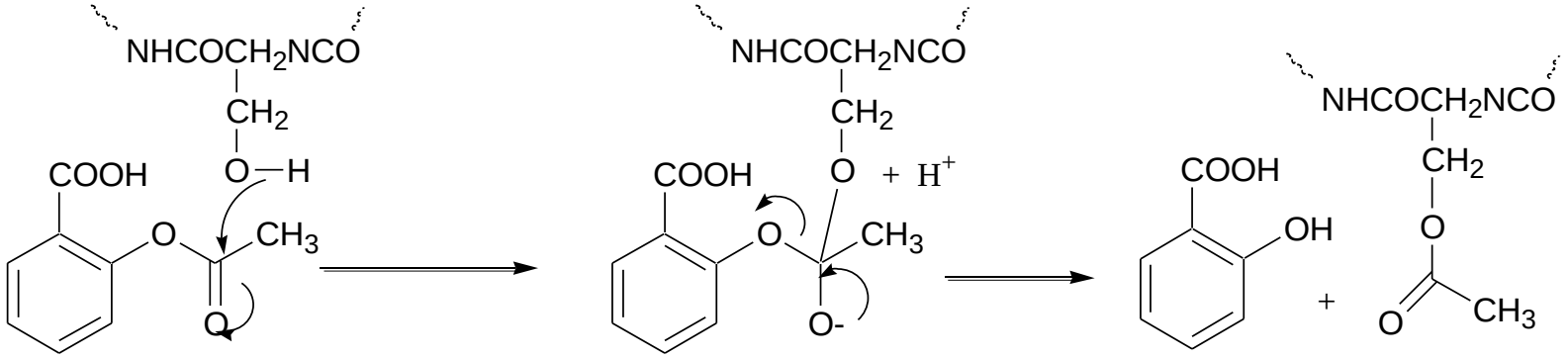
Aspirin



Acetylsalicylic acid

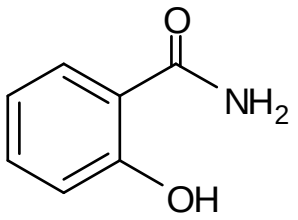
- The name **Aspirin** was coined by adding an ***a*** for acetyl to **spirin** from the name of the plant from which salicylic acid was first isolated
- Produced by acetylation of salicylic acid with acetic anhydride.
- Aspirin is indicated for:-
  - ✓ The relief of minor aches and mild to moderate pain
  - ✓ To reduce the risk of transient ischemic attacks
  - ✓ For myocardial infarction prophylaxis
  - ✓ For arthritis and related arthritic conditions

- It is slightly soluble in water
- Hydrolyzed rapidly by esterases
- ASA irreversibly inhibits the enzyme by acetylating a serine residue thus preventing access to the cyclooxygenase site



Salicylamide

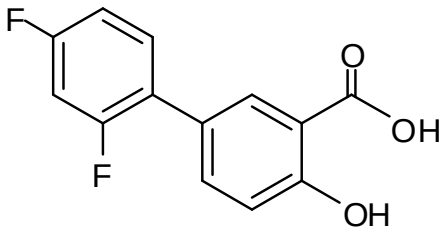
SALICYLAMIDE
(Bisalate)



- It is an isostere of salicylic acid
- OH replaced by NH2 to produce less acidic drug which is stable in aqueous preparations
- Does not cause GI tract ulceration and is absorbed only in intestine.
- It has greater CNS penetration.
- Effective as Aspirin as an analgesic/antipyretic and is effective in relieving arthritis pain
  - ✓ but does not appear to have anti-inflammatory actions.

Diflunisal

DIFLUNISAL
Dolobid

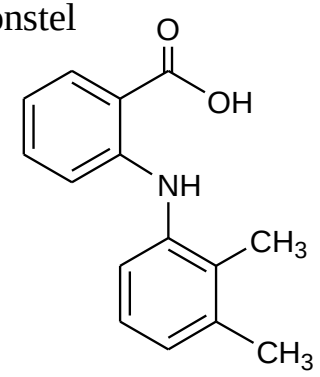


- Diflunisal is absorbed only from intestine;
- It is not soluble in gastric fluid.
  - ✓ Thus, gastric bleeding and GI upset is not as common
- Has increased duration of action.
  - ✓ It lasts 3–4 times longer than aspirin.
- It is moderately potent inhibitor of prostaglandin biosynthesis
  - ✓ Because it contain two aromatic ring

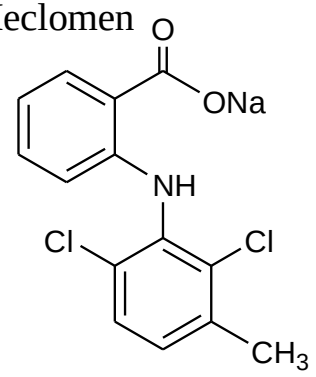
Fenemates

- The Fenemates are derivatives of Anthranilic acid, an isoster of salicylic acid
- The most potent analogs are those disubstituted at 2’ and 3’.
- This indicates that activity resides in compounds with the substituent on the second ring that keep it out of coplanarity by the *ortho* substituent

MEFENAMIC ACID
Ponstel

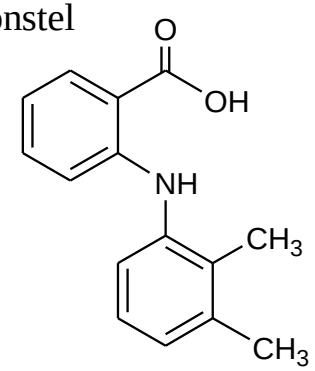


MECLOFENAMATE Sodium
Meclomen

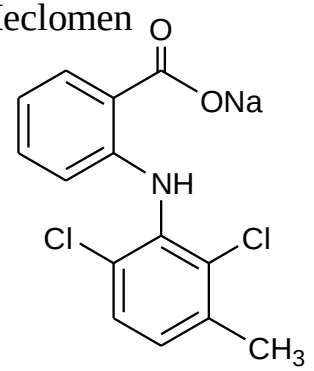


- **Mefenamic** acid has only one substituent, the 2’ methyl, that ensures non coplanarity
- **Meclofenamate** sodium has two such groups, the chlorine atoms, and thus more molecules of Meclofenamate assume the correct conformation and the drug is more potent
- Meclofenamate is 25 times more potent thus normal dose for Meclofenamate is 25 mg while the dose for Mefenamic acid is 250 mg.
- Since this class offers no advantage over the salicylates with respect to analgesic or anti-inflammatory actions, there is little interest in developing this class

MEFENAMIC ACID
Ponstel



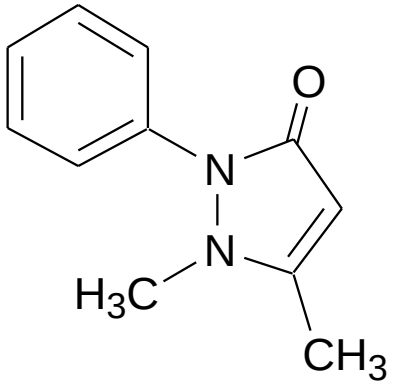
MECLOFENAMATE Sodium
Meclomen



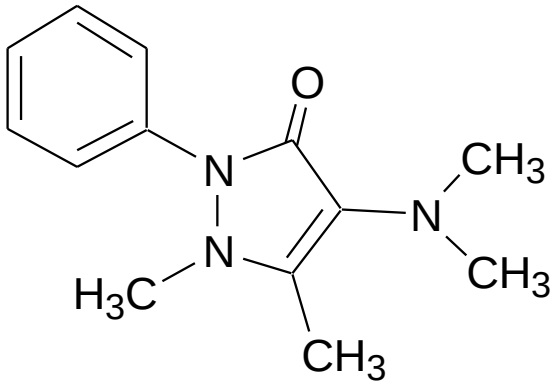
Pyrazoles and Pyrazolidinediones

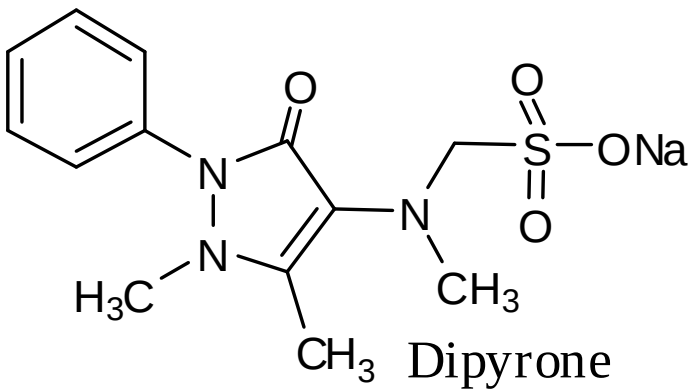
- **Antipyrine** is the prototype
  - ✓ Antipyretic and analgesic activities were discovered by accident.
- **Aminopyrine** is an analog,
  - ✓ More potent and longer acting
- Both possess significant incidences of agranulocytosis leading to death and used only in otic drops

ANTIPYRINE (Auralgan Otic)

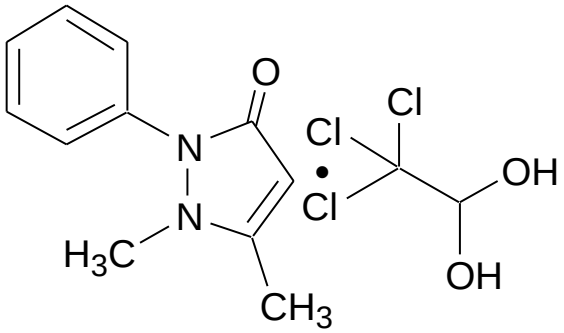


Aminopyrine



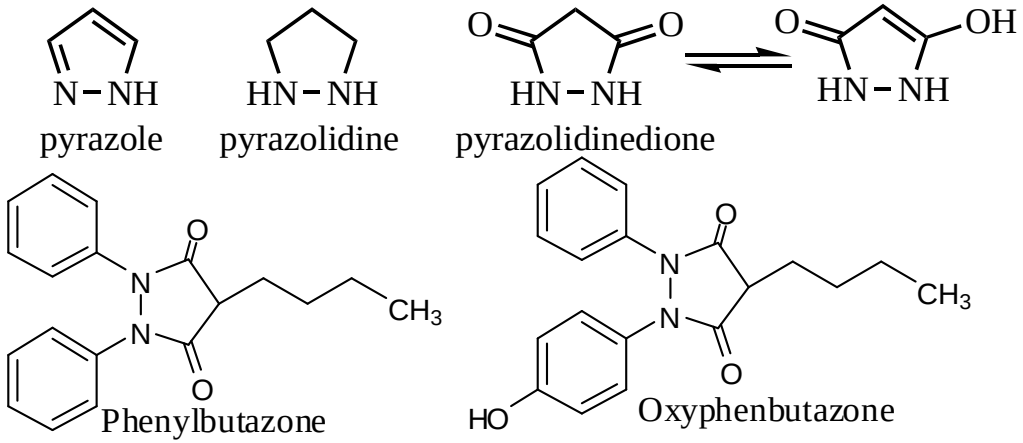


Dipyrone



DICHLORALPHENAZONE
Antipyrine • Chloral Hydrate

- **Dipyrone** is a prodrug
  - ✓ Spontaneously decomposes in aqueous solutions to aminopyrine.
- **Dichloralphenazone** is a complex of Aminopyrine and Chloral hydrate
  - ✓ It is a mild sedative used in migraine /tension headache products.



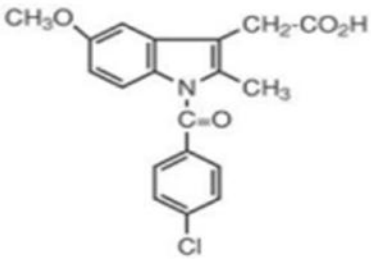
Phenylbutazone

- It is equipotent to antipyrine
- More potent than aspirin for treating inflammation.
- It has long half-life (72-84 hours).
- Serious **toxicities** limits its use in long term therapy.
  - ✓ e.g., agranuylocytosis, peptic ulcers and bone marrow depression,

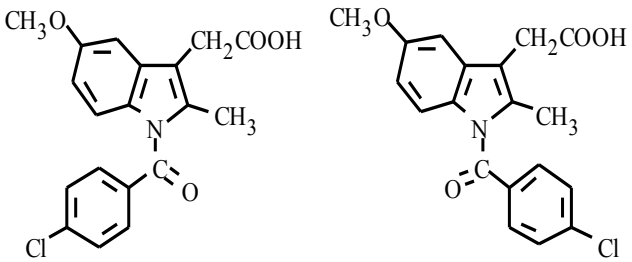
Oxyphenbutazone

- It's the active metabolite of Phenylbutazone with similar activity
- Equipotent but less toxic than phenylbutazone
- Shorter half–life (half-life 48-72 hours) and **better** tolerated.

Indomethacin

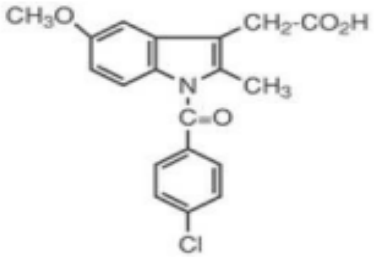


- It is more potent antipyretic than either aspirin or acetaminophen
- 10 times more potent analgesic than aspirin.



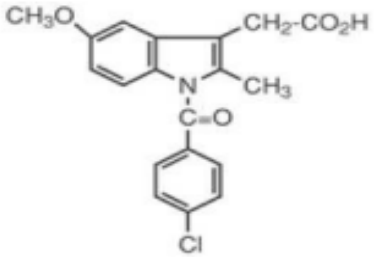
Different conformers of Indomethacin

SAR



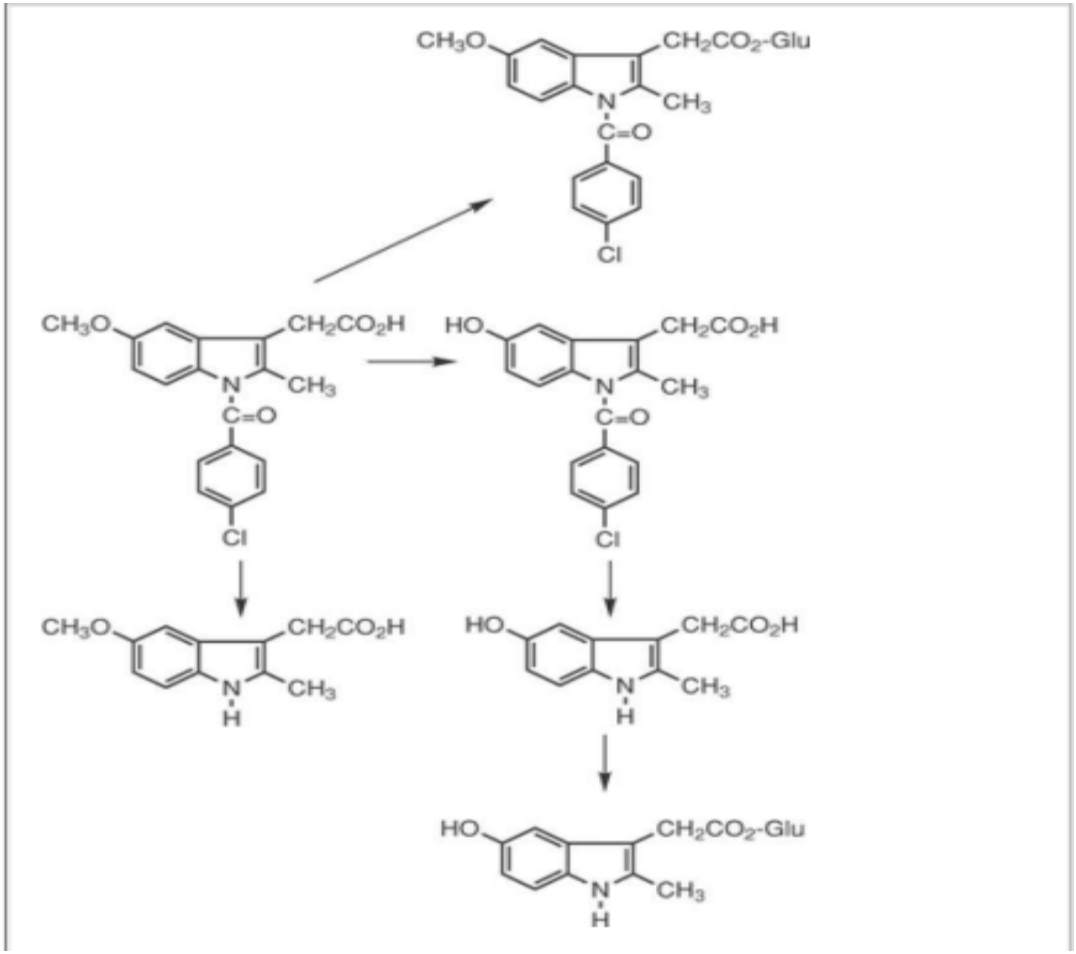
- Replacement of the carboxyl group with other acidic functionalities decreases activity.
  - ✓ Anti- inflammatory activity generally increases as the acidity of the carboxyl group increases and decreases as the acidity is decreased
- Amide analogues are inactive
- Acylation of the indole nitrogen with aliphatic carboxylic acids
  - ✓ Results in amide derivatives that are less active than those derived from benzoic acid

Cont…

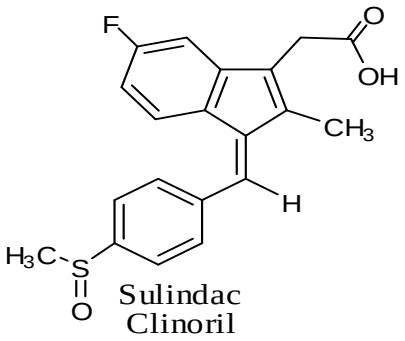


- The presence of an indole ring nitrogen is not essential for activity
  - ✓ Because the corresponding 1- benzylidenylindene analogues (e.g., sulindac) are active.
- Alkyl groups, especially methyl, at the α-position are more active than aryl substituents
- 5-position of the indole ring is most flexible with regard to the nature of substituents that enhance activity.
  - Substituents such as methoxy, fluoro, dimethylamino, methyl and acetyl are more active than the unsubstituted indole ring

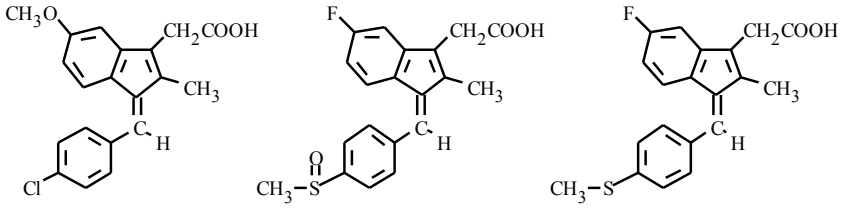
Metabolism by glucuronidation



Sulindac



- Indomethacin has significant CNS side effects due to the indole nucleus.
  - ✓ Thus the heterocyclic nitrogen was removed and a double bond introduced, giving the *indene* derivative.
- Z isomer is active,
- Lacks the CNS side effects and causes less GI irritation but low water solubility.
  - ✓ Which is alleviated by replacing the chlorine atom of the phenyl substituent with a sulfinyl group
- The 5-fluorine is used to enhance analgesic activity



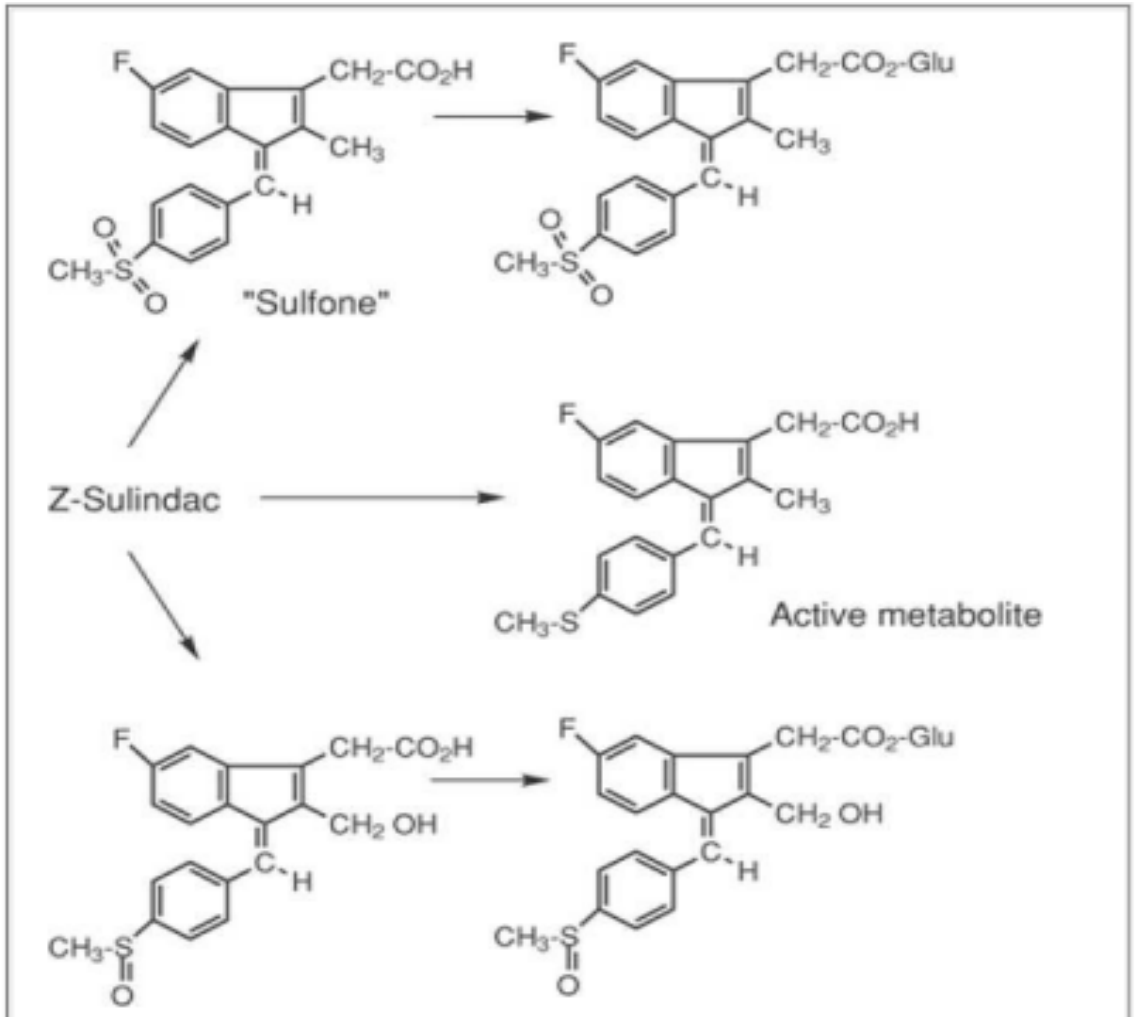
Indene analog of Indomethacin | Sulindac | Sulfide of Sulindac

- Sulindac is a prodrug.
- Its active form is the sulfide metabolite which has a long half–life allowing for BID administration.
- Clinically it has only about half the potency of Indomethacin in treating inflammation and reducing fever.
  - ✓ But; it is equipotent in analgesic effect.
- Since the drug is absorbed as the inactive sulfoxide,
  - ✓ It causes fewer GI disturbances (no prostaglandin biosynthesis inhibition in the stomach).
- The thioether metabolite (shown at the right) is longer-lived than the parent

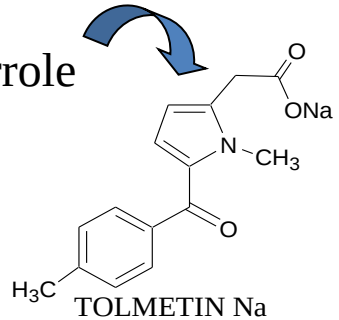
Metabolism

- Once sulindac enters the circulatory system, it is reduced to the sulfide
  - ✓ Which is an inhibitor of prostaglandin biosynthesis in the joints.
- The active metabolite has a plasma half-life approximately twice that of the parent compound
- In addition to the sulfide metabolite, sulindac is oxidized to the corresponding sulfone
  - ✓ which is inactive.
- A minor product results from hydroxylation of the benzylidene function and the methyl group at the 2-position.
- Glucuronides of several metabolites also are found

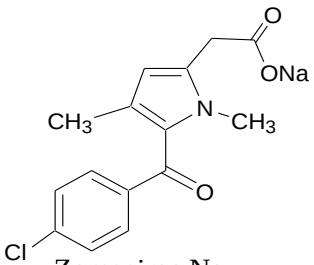
Metabolism


F
CH2-CO2H
CH3
C
H
O
CH3-S
O
"Sulfone"
CH2-CO2-Glu
Z-Sulindac
Active metabolite
CH2 OH

Indole replaced with pyrrole



TOLMETIN Na
Tolectin



Zomepirac Na
[Zomax]

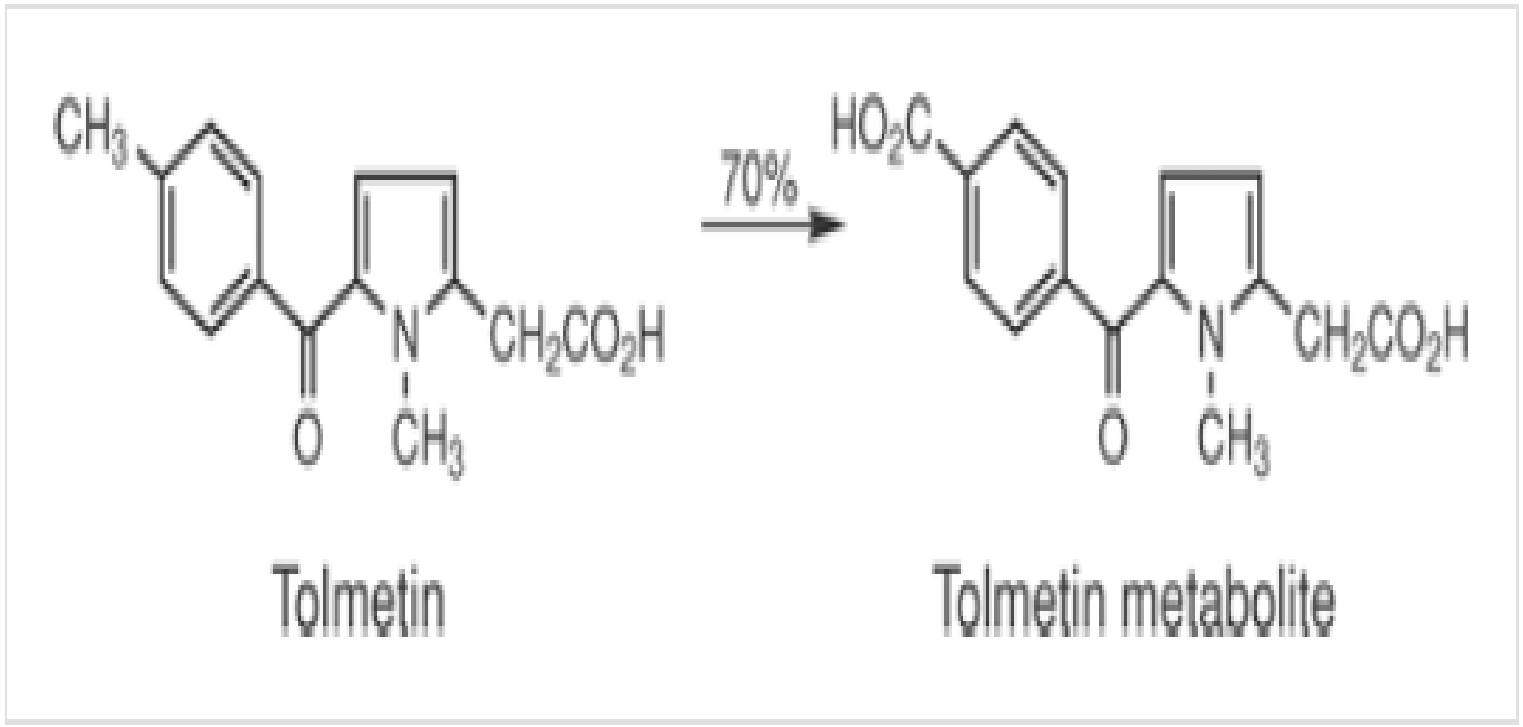
Tolmetin

- Contain the three portions of Indomethacin deemed necessary for activity, the carboxyl, the flat indole ring and an out of plane phenyl
- The major metabolite is the carboxylic acid resulting from benzylic hydroxylation and subsequent oxidation.
- It has a half–life of 30 to 60 minutes
- To increase the duration of action
  - ✓ The methyl was replaced with a chloro which prevented metabolism at the phenyl ring.
    - This drug was **Zomepirac** which was marketed but eventually removed due to reports of severe anaphylactic reactions in patients sensitive to Aspirin.

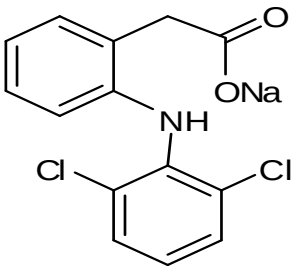
Cont..

- Inhibits prostaglandin biosynthesis.
- Tolmetin, however, also inhibits polymorph migration and decreases capillary permeability.

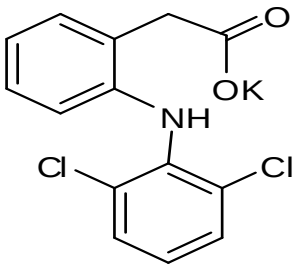
Metabolism


CH3
N
CH2CO2H
O
CH3
70%
HO2C
N
CH2CO2H
O
CH3
Tolmetin
Tolmetin metabolite

Diclofenac



Diclofenac Na
Voltaren, (Arthrotec), Solaraze



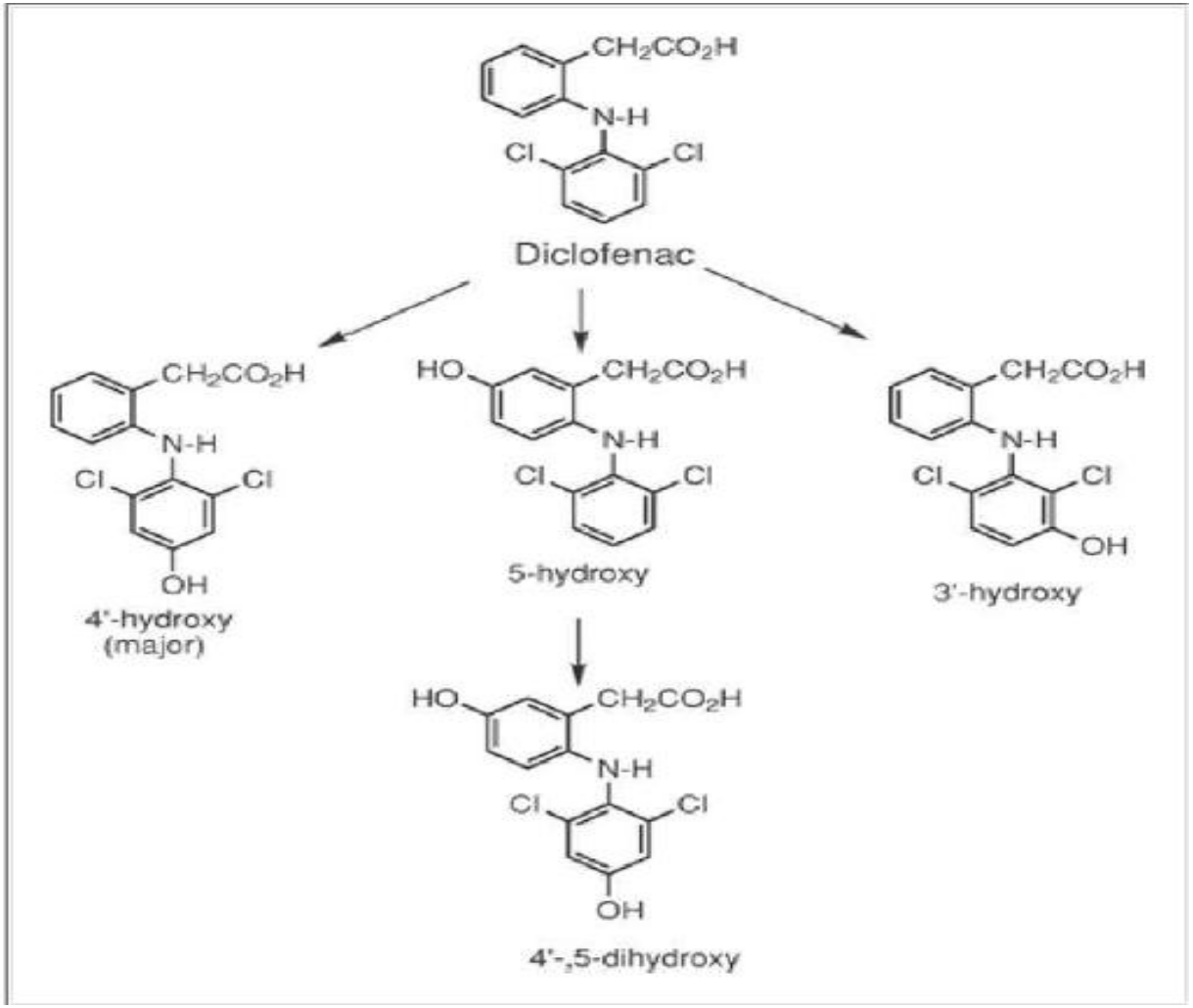
Diclofenac K
Cataflam

- Is the most popular NSAID.
- Its mechanism of action may be a little different from the others.
  - ✓ It is a COX inhibitor like the rest, but it also seems to inhibit *lipoxygenase* to some degree.
    - ❖ This could account for its increased anti-inflammatory effectiveness and potency.
- The two Cl groups are necessary to force the two rings out of plane with each other.
- It has a profile of action similar to the others and favors anti-inflammation uses, rather than analgesic uses.

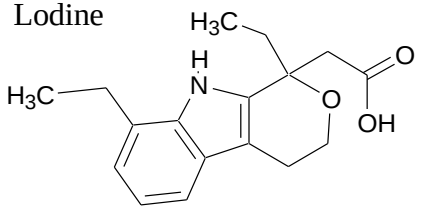
Metabolism

- Only 50 to 60% of an oral dose is bioavailable because of extensive hepatic metabolism
- Four major metabolites resulting from aromatic hydroxylation have been identified
- The major metabolite via CYP3A4 is the 4′-hydroxy derivative and accounts for 20 to 30% of the dose excreted
- The 5-hydroxy, 3′-hydroxy, and 4′,5-dihydroxy metabolites via CYP2C9 account for 10 to 20% of the excreted dose.
- The remaining drug is excreted in the form of sulfate conjugates.
- The major metabolite is much less active than the parent compound

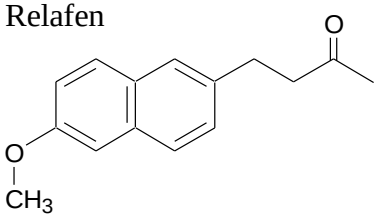
Metabolism


CH2CO2H
N-H
Cl
Cl
Diclofenac
HO
OH
4'-hydroxy
(major)
5-hydroxy
3'-hydroxy
4'-,5-dihydroxy

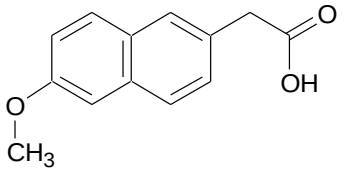
ETODOLAC
Lodine



NAMBUMETONE
Relafen



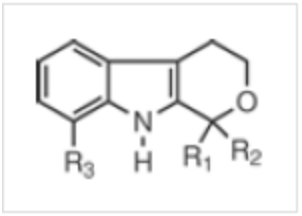
6–Methoxy–2–naphthylic acid
6–MNA



Etodolac

- Can be considered a nonclassical bioisostere of the arylpropionic acids.
- It is:-
  - ✓ 50x more potent than aspirin in inflammation.
  - ✓ 33% as potent as indomethacin.
  - ✓ It shows a much better GI profile than aspirin or indomethacin and.

SAR

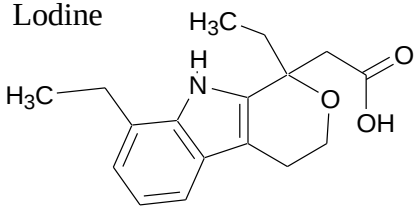


- Alkyl groups at R1 and an acetic acid function at R2 enhanced anti-inflammatory activity.
  - ✓ Lengthening the acid chain, or ester or amide derivatives, gave inactive compounds.
  - ✓ The corresponding α-methylacetic acid derivatives also were inactive.
- Increasing the chain length of the R1 substituent to ethyl or n-propyl gave derivatives that were 20 times more potent than methyl

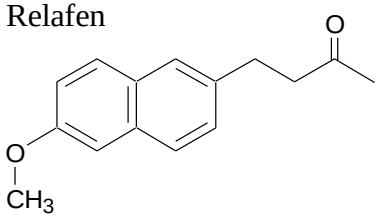
Cont…

- A number of aromatic substituents in the aromatic ring were evaluated
  - ✓ Substituents at the 8-position were most beneficial .
    - Among the most active were the 8-ethyl, 8-n-propyl , and 7-fluoro-8-methyl derivatives.

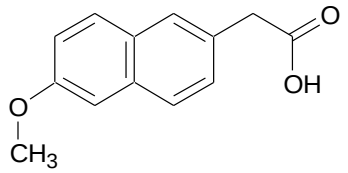
ETODOLAC
Lodine



NAMBUMETONE
Relafen



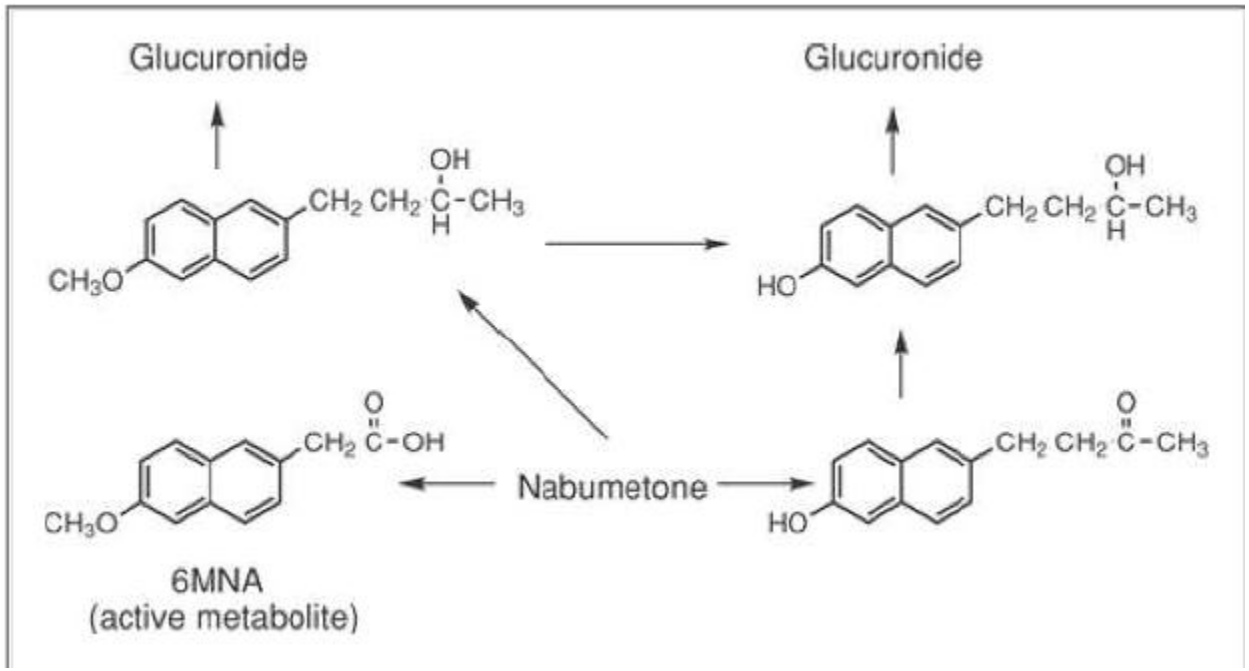
6–Methoxy–2–naphthylic acid
6–MNA



Nabumetone

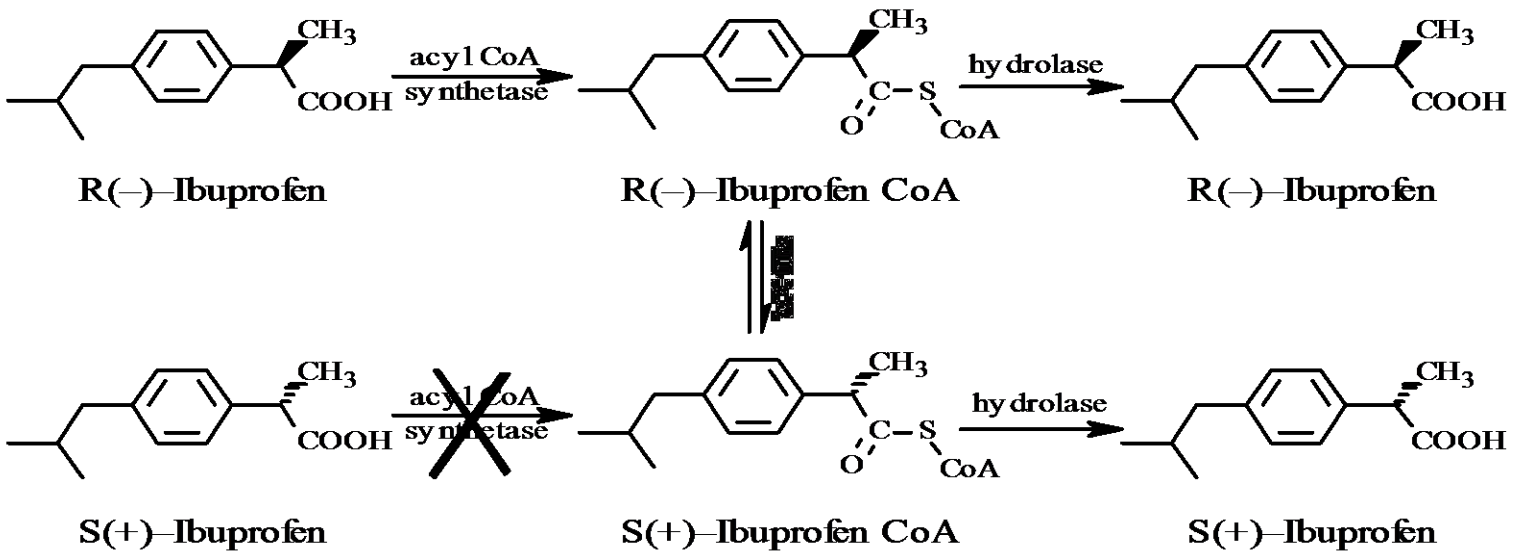
- It is a ketone and thus non–acidic pro-drug.
  - ✓ Does not produce a significant primary insult
- It is classed as an **Alkanone.**
- It is a prodrug and must be activated by metabolizing in to 6–Methoxy–2–naphthylacetic acid.
- Approximately 35% of a 1000mg dose is converted to 6–MNA, which is structurally related to Naproxen

Metabolism

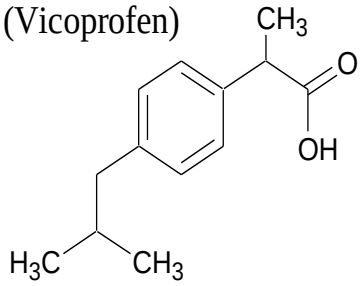

Glucuronide
Glucuronide
OH
CH2 CH2 C-CH3
H
CH3O
HO
O
CH2 C-OH
CH2 CH2 C-CH3
Nabumetone
6MNA
(active metabolite)

Arylpropioanic acid Derivatives

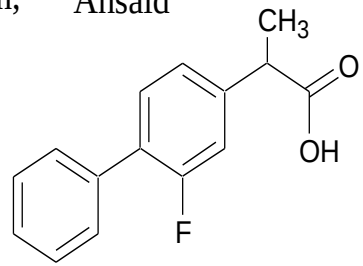
- They have chiral center
  - ✓ Activity resides in the S isomer.
- ***in vivo*** some of the inactive R isomer is converted to the active S by isomerases
  - but not the S to R.
- One reference states that 60% of an Ibuprofen and 100% of a Fenoprofen does undergo isomerization.



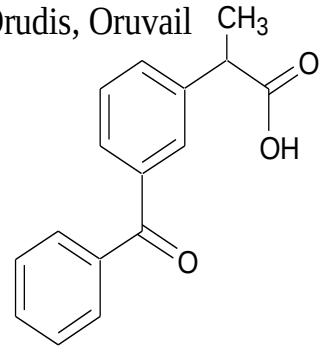
IBUPROFEN
Motrin, Rufen, Advil, Nuprin, (Vicoprofen)



FLURBIPROFEN
Ansaid



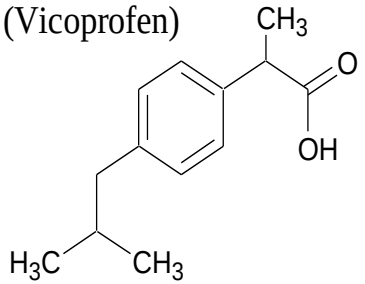
KETOPROFEN
Orudis, Oruvail



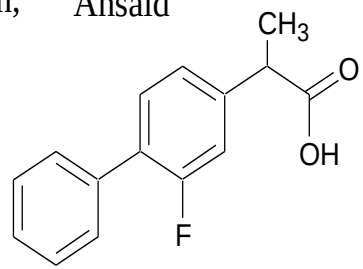
Ibuprofen

- Marketed as the racemate.
  - ✓ Biological activity resides almost exclusively in the *S*-(+)-isomer.
- Ibuprofen is more potent than aspirin but less potent than indomethacin in anti-inflammatory
- It produces moderate degrees of gastric irritation.

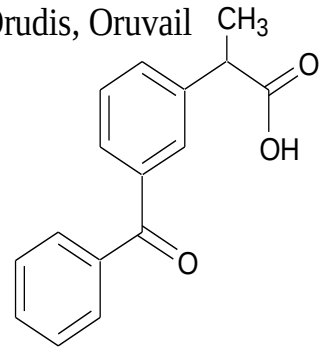
IBUPROFEN
Motrin, Rufen, Advil, Nuprin, (Vicoprofen)



FLURBIPROFEN
Ansaid



KETOPROFEN
Orudis, Oruvail



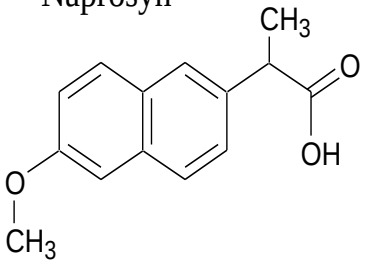
Flurbiprofen

- Resulted from a study of the SARs.
  - ✓ The 3–fluoro substituent helps ensure non–coplanarity.
- Introduced as a topical product for ophthalmic use (Ocufen).
  - ✓ Later it was introduced for systemic use
- This drug is many times the potency of the other drugs
  - ✓ 100x phenylbutazone against inflammation, and
  - ✓ Is about half as potent as methylprednisolone (an anti-inflammatory steroid).

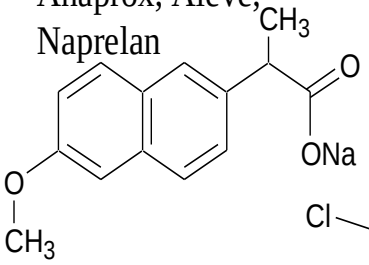
Ketoprofen (1986):

- In addition to arachidonic acid pathway, it also inhibits the leukotriene pathway
- It is clinically less potent than Indomethacin, but has about the same GI disturbance profile

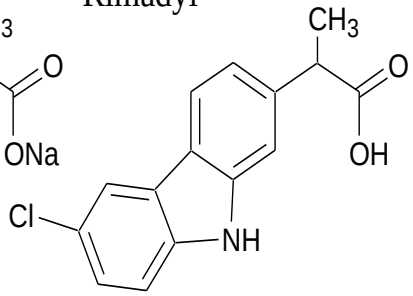
NAPROXEN
Naprosyn



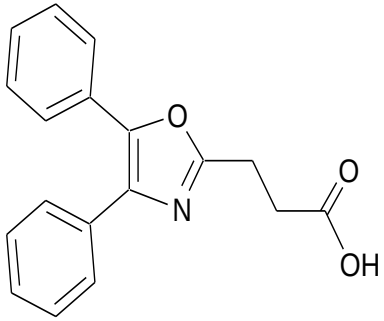
NAPROXEN Sodium
Anaprox, Aleve, Naprelan



CARPROFEN
Rimadyl



OXAPROZIN
Daypro



Naproxen

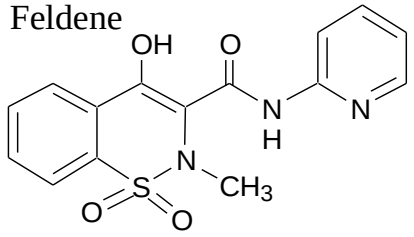
- Does not possess a second non coplanar ring
- It is marketed as the *S*-(+)-enantiomer, but interestingly, the sodium salt of the (–)-isomer also is on the market as Anaprox.
- As an inhibitor of prostaglandin biosynthesis,
  - ✓ It is 12 times more potent than aspirin,
  - ✓ 10 times more potent than phenylbutazone,
  - ✓ Three to four times more potent than ibuprofen, and
  - ✓ But, 300 times less potent than indomethacin

Cont…

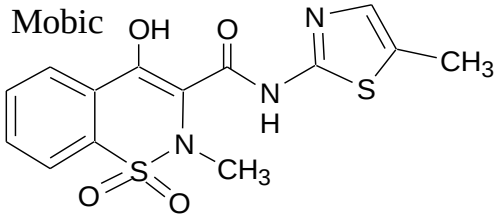
- Replacing the carboxyl group with functional groups capable of being metabolized to the carboxyl function (e.g., —CO_2CH_3, —CHO, or —CH_2OH) led to a retention of activity.
- Naproxen is the only NSAID currently marketed as optically active isomers of this class
- **Carprofen** is marketed as a veterinary analgesic.
- **Oxaprozin** is an aryl propionic acid but is unique in that the propyl is not branched.

Oxicams

PIROXICAM
Feldene



MELOXICAM
Mobic

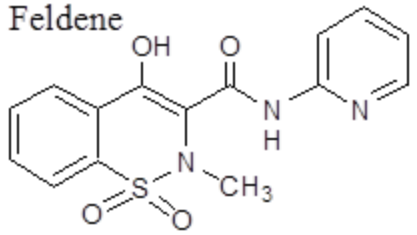


- Pfizer developed this class to produce non–carboxylic acid NSAIDS in an effort to produce, potent, and well -tolerated anti-inflammatory drugs.

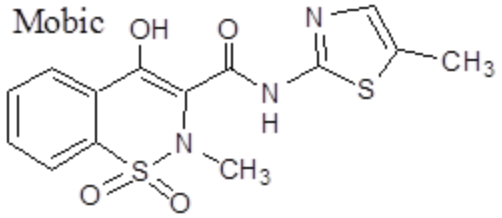
Piroxicam

- Is the first member of this family marketed
- Although it has good potency
  - ✓ The GI side effects limit its usefulness.
- A typical half-life for Piroxicam is 38 hours.

PIROXICAM
Feldene



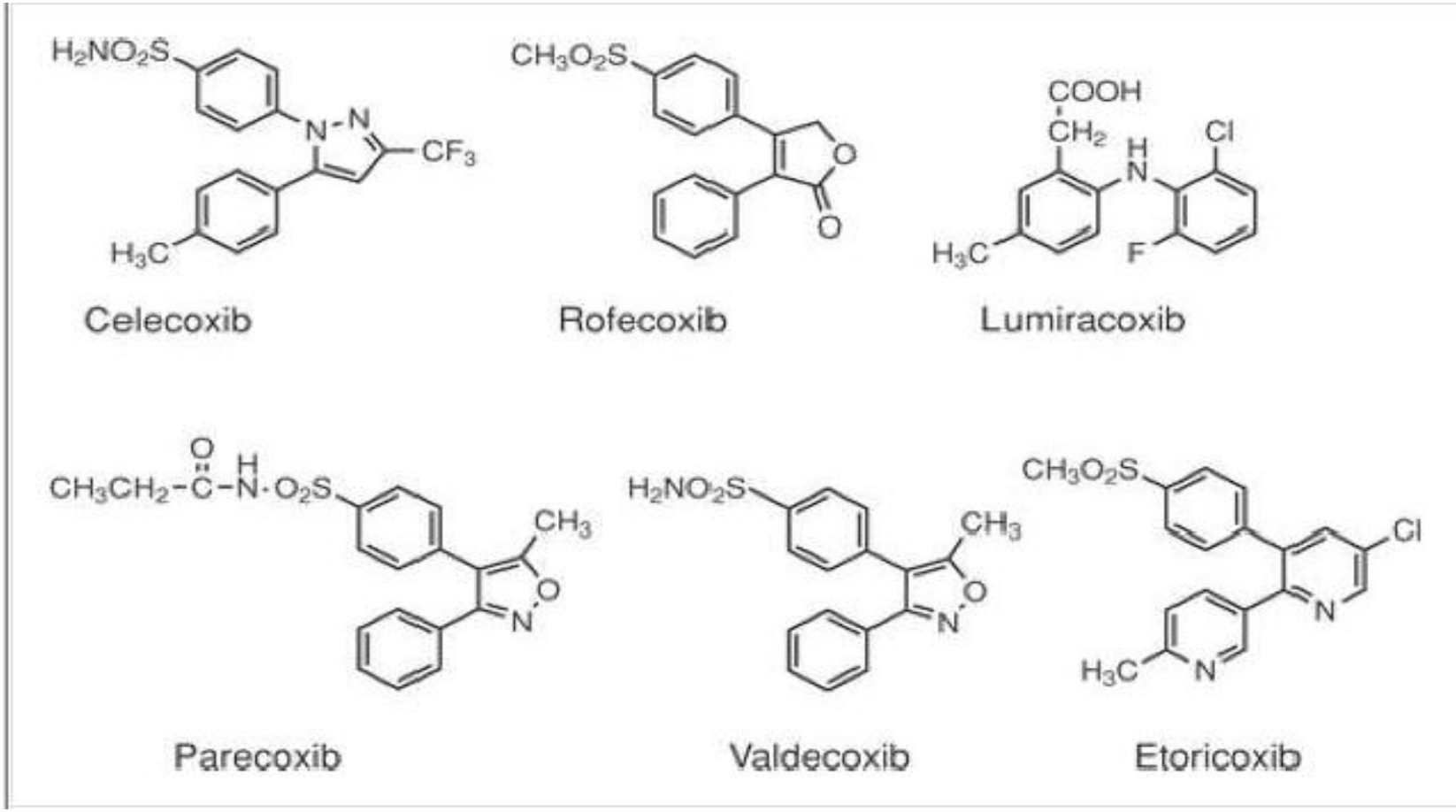
MELOXICAM
Mobic



- **Meloxicam** Is structurally related to Piroxicam.

- Although Meloxicam is frequently described in the literature as a selective COX-2 inhibitor, but.
  - ✓ It is considerably less selective for the COX-2 versus COX-1 isoenzyme when compared to Celecoxib or Rofecoxib.

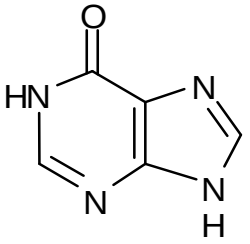
Selective COX-2 Inhibitors



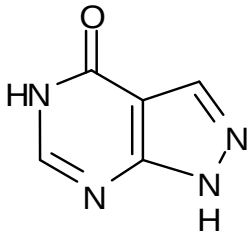
- Structurally they differs from other NSAIDS in that they are only weakly acidic.
- It does possess a sulfamyl group and has a warning about use in patients with a sulfonamide allergy
- **Valecoxib** is also a sulfamyl and its package insert contains the same caution.
  - ✓ It is this phenyl group which is inserted into the extra space in COX–2
- **Deracoxib** is also acidic but is indicated for veterinary use
- **Rofecoxib** is not acidic

Allopurinol

- Is a structural analog of hypoxanthine and
  - ✓ Thus is a xanthine oxidase inhibitor used to treat hyperuricemia and its complications including chronic gout
  - ✓ prophylaxis with chemotherapeutic treatments, which can rapidly produce severe hyperuricemia.



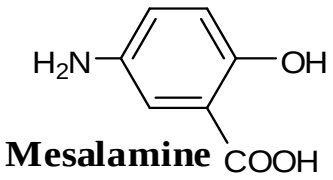
Hypoxanthene



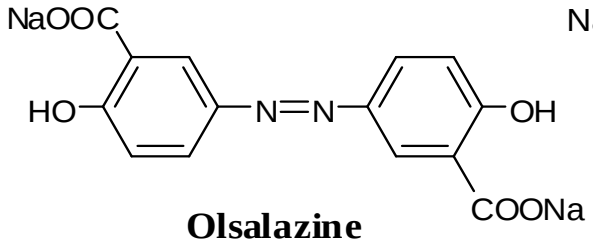
Allopurinol

Drugs for Inflammatory Bowel Disease (Ulcerative Colitis)

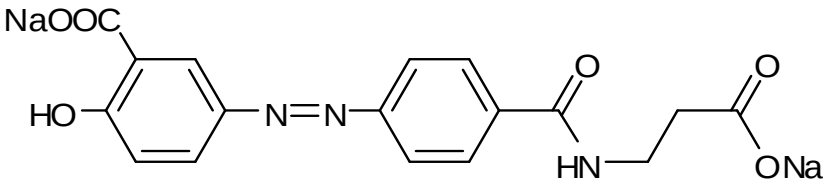
- Mesalamine or 5-aminosalicylic acid (5-ASA), and its prodrugs balsalazine and olsalazine are anti-inflammatory drugs/prodrugs used to treat inflammation of the digestive tract (ulcerative colitis) and mild-to-moderate Crohn's disease.
- Mesalazine is a bowel-specific aminosalicylate drug that acts locally in the gut and has its predominant actions there, thereby having few systemic side effects.
- Balsalazine and olsalazine generate mesalamine in the site of action.



Mesalamine



Olsalazine



Balsalazine

The end!

Any comment, question…